Fiscal Backing and Heterogeneous Household Portfolios*

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Abstract

When do fiscal deficits become inflationary? I answer this question in a heterogeneous-agent New Keynesian model in which households hold a liquid portfolio of nominal government liabilities and tradable profit claims, alongside illiquid real capital. The answer depends on fiscal backing. Deficits that are not backed by higher expected future primary surpluses generate much larger inflation and output responses than backed deficits, because weak backing induces households to reduce their demand for nominal government liabilities and rebalance toward real capital. The increment in impact consumption associated with unfunded policy is similar across government purchases and transfers, although their response levels differ. In the reduced first-order solution, more frequent adjustment of illiquid assets lowers impact consumption under funded transfers and raises it under unfunded transfers. The results show that portfolio choice makes fiscal backing quantitatively central for inflation.

JEL codes: E62, E31, E52, D31, G11

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Introduction

Large fiscal expansions and elevated inflation after the COVID-19 pandemic have renewed interest in a basic question: under what circumstances do fiscal deficits raise inflation? A fiscal deficit does not, by itself, imply inflation. When higher government debt is expected to be matched by higher future primary surpluses, the fiscal expansion mainly changes the timing of fiscal adjustment rather than the valuation of government debt. Fiscal explanations of inflation require that deficits have real effects that are not neutralized by expected future surpluses.

One channel comes from the Fiscal Theory of the Price Level (FTPL). The real value of outstanding nominal government debt must be consistent with expected future fiscal backing. When higher deficits are not expected to be matched by higher future primary surpluses, the price level rises to reduce the real value of nominal government debt until household asset demand and fiscal backing are again consistent with equilibrium (Cochrane, 2001; Leeper, 1991; Woodford, 1995). This logic has been used to interpret both earlier episodes of fiscally driven inflation and the post-COVID inflation surge (Balke and Zarazaga, 2024; Barro, 2022; Bianchi et al., 2023; Bianchi and Ilut, 2017; Hall and Sargent, 2011).

An alternative channel comes from heterogeneous-household models with nominal rigidities. Fiscal deficits become inflationary when they shift resources toward households with high marginal propensities to consume (MPCs), whose consumption responds strongly to additional income. The resulting rise in aggregate demand generates quantitatively important inflation responses even without making fiscal backing central to the analysis (Angeletos et al., 2024). This literature shows that household heterogeneity can account for substantial inflationary effects of fiscal deficits. That conclusion, however, is derived in environments in which households save in a single liquid nominal government bond and do not choose whether to reallocate wealth into illiquid real capital.

I ask how the inflationary effects of fiscal deficits change once households can also shift wealth between liquid and illiquid real assets. In environments with only a liquid nominal

government asset, households can respond to weak fiscal backing only by changing consumption and nominal saving. In this paper, they can also rebalance away from nominal government liabilities and toward real capital. This additional margin is central because it changes how fiscal backing is transmitted to inflation and real activity.

Building on the framework in Bayer et al. (2024), I develop a heterogeneous-agent New Keynesian (HANK) model with long-term nominal government debt, sticky prices and wages, and endogenous household portfolio choice. Households can save or borrow through a liquid portfolio that includes government liabilities and tradable profit claims, but can also hold illiquid real capital that can be adjusted only intermittently. Fiscal policy operates through government purchases and transfers, including both uniform lump-sum transfers and transfers targeted to different household groups. I compare funded deficits, which are backed by higher expected future primary surpluses, with unfunded deficits, for which future primary surpluses do not provide sufficient backing.

The main findings are fivefold. First, unfunded deficits generate much larger aggregate responses than funded deficits. When future primary surpluses do not provide sufficient fiscal backing, households reduce their demand for nominal government liabilities and rebalance their portfolios toward illiquid real capital. This portfolio reallocation accompanies the decline in the real value of nominal government debt and the larger responses of investment, output, and inflation. Second, the increase in impact consumption from changing fiscal backing is very similar across government purchases and transfers, although the levels of their responses differ. This result suggests that fiscal backing is an important determinant of aggregate responses alongside the fiscal instrument. Third, unfunded transfers lower consumption inequality and initially raise consumption most strongly in the bottom and middle of the liquid-wealth distribution; consumption at the top also rises initially but falls below baseline from quarter five onward. The decline in the real value of nominal government debt reduces the wealth of households with large nominal asset positions, while higher output and transfers support households with lower wealth. Fourth, when deficits are

funded, the type of fiscal expansion matters. Because funded deficits are backed by higher expected future primary surpluses, they do not trigger a large portfolio reallocation away from nominal government bonds to the same extent as unfunded deficits do. Redistribution through the fiscal instrument becomes more important: funded transfers provide immediate support to their recipients, while government purchases enter goods demand directly.

Fifth, lessening the frictions that restrict households' access to the illiquid capital market changes the quantitative gap between the responses to funded and unfunded deficits. Moving from $\lambda = 0.005$ to the baseline $\lambda = 0.03$ lowers funded impact consumption and raises unfunded impact consumption in the reduced first-order solution. A decline in asset-holding frictions has two effects. First, households can self-insure idiosyncratic risk more effectively and smooth consumption more over time. Second, households can rebalance wealth away from liquid assets more easily when fiscal backing weakens. The net effect is quantitative: in this comparison, the difference between unfunded and funded consumption responses widens. These experiments compare separately computed stationary economies and do not isolate either margin on its own.

This paper contributes to two primary strands of literature. First, it speaks to the growing literature on fiscal deficits, inflation, and the FTPL in heterogeneous-agent environments. The FTPL literature shows that the real value of nominal government debt depends on expected fiscal backing (Bianchi et al., 2023; Bianchi and Ilut, 2017; Cochrane, 2001; Hall and Sargent, 2011; Leeper, 1991; Woodford, 1995). Closest in spirit, Leeper et al. (2017) study fiscal financing and monetary-fiscal regimes in an estimated DSGE model with savers and non-savers, focusing on government spending multipliers across regimes. Recent work brings fiscal backing into richer heterogeneous-agent environments: Kaplan et al. (2026) develop a heterogeneous-agent fiscal theory of the price level, but in a flexible-price economy in which households save in a single liquid nominal government asset. Other recent work emphasizes a redistribution channel: deficits can raise inflation by shifting resources toward households with high marginal propensities to consume under nominal rigidities (Angeletos

et al., 2024; Kwicklis, 2025). These papers show that household heterogeneity matters for fiscal inflation, but they do not study how fiscal backing is transmitted when households can choose between liquid nominal government liabilities and illiquid real capital.

In the current paper, I study that portfolio-reallocation margin. Once households can choose between a liquid portfolio and real capital, redistribution and portfolio adjustment jointly shape the inflationary effects of fiscal deficits. When deficits are unfunded, weak fiscal backing induces households to reduce their demand for nominal government liabilities and shift wealth toward real assets. Unfunded deficits are much more inflationary than funded deficits in the calibration, and the increment in impact consumption associated with fiscal backing is similar across instruments. The comparison highlights a role for fiscal backing alongside MPC heterogeneity and fiscal targeting. Whether the deficit is fiscally backed becomes the central determinant of the aggregate response to fiscal shocks.

Second, I contribute to the literature on household portfolios and policy transmission. Quantitative work with liquid and illiquid assets shows that household portfolio structure matters for aggregate dynamics and policy transmission (Bayer et al., 2023, 2024; Luetticke, 2021). I use this liquid-illiquid portfolio structure to study when fiscal deficits become inflationary. Lowering frictions in the illiquid capital market changes the difference between funded and unfunded deficits: in the reduced first-order, one-time-transfer comparison, impact consumption falls under funded policy and rises under unfunded policy as adjustment becomes more frequent. This result shows that household heterogeneity and portfolio choice do not replace the FTPL explanation of inflation. Instead, portfolio choice shapes how households respond when fiscal backing weakens.

The remainder of the paper is organized as follows. Section 1 presents the model. Section 2 discusses the calibration. Section 3 reports the quantitative results. Section 4 concludes.

1 Model

I build a heterogeneous-agent New Keynesian model with incomplete markets, long-term nominal government debt, and endogenous household portfolio choice. Households save in two assets: a liquid portfolio containing nominal government liabilities and tradable profit claims, and illiquid real capital. The government runs deficits through government purchases and transfers and distinguishes between funded deficits, for which expected future primary surpluses provide sufficient backing, and unfunded deficits, for which expected future primary surpluses do not offset current deficits. Transfers may be distributed uniformly across households or targeted to specific household groups. Firms operate under monopolistic competition with sticky prices and wages. Banks intermediate household liquid assets into long-term nominal government debt.

The household side builds on Bayer et al. (2024), the banking and long-term debt side follows Bayer et al. (2023), and the funded-versus-unfunded fiscal structure follows Balke and Zarazaga (2024). Appendix A provides additional details on preferences, taxation, firm-side optimization, and wage setting.

1.1 Government and Fiscal Backing

The government issues long-term nominal debt, levies taxes on labor and profit income, and runs deficits through government purchases and transfers. I distinguish between funded government purchases and transfers, for which expected future primary surpluses provide sufficient backing, and unfunded government purchases and transfers, for which expected future primary surpluses do not provide sufficient backing. Monetary policy follows a passive Taylor rule.

1.1.1 Fiscal Authority

Let $Bgov_{t+1}$ denote the real market value of total government debt at the end of period t. I decompose total government debt into funded debt, $BFgov_{t+1}$, and unfunded debt, $BUgov_{t+1}$:

$$Bgov_{t+1} = BFgov_{t+1} + BUgov_{t+1}. \tag{1}$$

Total government debt evolves according to

$$Bgov_{t+1} = G_t + Transfer_t - Tax_t + \frac{\mathcal{R}_t^B}{\pi_t} Bgov_t, \tag{2}$$

where G_t is total government consumption, $Transfer_t$ is total government transfers, Tax_t is tax revenue, π_t is gross inflation, and $\mathcal{R}_t^B = [1 + (1 - \delta_B)q_t^B]/q_{t-1}^B$ is the realized nominal holding return on long-term debt. It includes bond-price changes and need not equal the predetermined short nominal rate R_t^b.

Funded debt evolves according to

$$BFgov_{t+1} = GF_t + TF_t - \chi_1 Tax_t + \frac{\mathcal{R}_t^B}{\pi_t} BFgov_t, \tag{3}$$

where GF_t and TF_t denote the funded components of government purchases and transfers, and $\chi_1 \in [0, 1]$ is the share of tax revenue allocated to the funded debt component. I set $\chi_1 = \bar{BFgov}/\bar{Bgov}$ in steady state. The unfunded component is then

$$BUgov_{t+1} = Bgov_{t+1} - BFgov_{t+1}. \tag{4}$$

Government purchases and transfers are split into funded and unfunded components:

$$G_t = GF_t + GU_t, \tag{5}$$

$$Transfer_t = TF_t + TU_t. \tag{6}$$

For each fiscal component $X \in \{GU, GF, TU, TF\}$, write $X_t = X_t^{\text{rule}} + D_t^X$, where D_t^X is the direct intervention. Out of steady state, the four rule components evolve according to

$$\frac{GU_t^{\text{rule}}}{\bar{GU}} = \left(\frac{GU_{t-1}^{\text{rule}}}{\bar{GU}}\right)^{\rho_{GU}} \left(\frac{Y_t}{Y_{t-1}}\right)^{(1-\rho_{GU})\gamma_{YGU}} \left(\frac{BUgov_t}{B\bar{U}gov}\right)^{(1-\rho_{GU})\gamma_{BGU}} \epsilon_t^{GU}, \tag{7}$$

$$\frac{GF_t^{\text{rule}}}{\bar{GF}} = \left(\frac{GF_{t-1}^{\text{rule}}}{\bar{GF}}\right)^{\rho_{GF}} \left(\frac{Y_t}{Y_{t-1}}\right)^{(1-\rho_{GF})\gamma_{YGF}} \left(\frac{BFgov_t}{B\bar{F}gov}\right)^{(1-\rho_{GF})\gamma_{BGF}} \epsilon_t^{GF}, \tag{8}$$

$$\frac{TU_t^{\text{rule}}}{\bar{TU}} = \left(\frac{TU_{t-1}^{\text{rule}}}{\bar{TU}}\right)^{\rho_{TU}} \left(\frac{Y_t}{Y_{t-1}}\right)^{(1-\rho_{TU})\gamma_{YTU}} \left(\frac{BUgov_t}{B\bar{U}gov}\right)^{(1-\rho_{TU})\gamma_{BTU}} \epsilon_t^{TU}, \tag{9}$$

$$\frac{TF_t^{\text{rule}}}{\bar{TF}} = \left(\frac{TF_{t-1}^{\text{rule}}}{\bar{TF}}\right)^{\rho_{TF}} \left(\frac{Y_t}{Y_{t-1}}\right)^{(1-\rho_{TF})\gamma_{YTF}} \left(\frac{BFgov_t}{B\bar{F}gov}\right)^{(1-\rho_{TF})\gamma_{BTF}} \epsilon_t^{TF}. \tag{10}$$

The parameters $\gamma_{BG_}$ and $\gamma_{BT_}$ measure how strongly government purchases and transfers respond to debt. A higher value implies that higher debt is followed by a larger increase in the corresponding spending component. This makes it harder for future primary surpluses to rise enough to provide sufficient backing for the debt. In the calibration, the funded coefficients are $\gamma_{BGF} = \gamma_{BTF} = -0.50$, while both unfunded coefficients are zero, $\gamma_{BGU} = \gamma_{BTU} = 0$. Higher funded debt therefore induces spending adjustment; neither unfunded component responds directly to debt. Tax revenues and output-related fiscal adjustments remain endogenous.

The benchmark intervention is a one-time additive payment: $D_0^{TF} = d$ for funded transfers or $D_0^{TU} = d$ for unfunded transfers, with $D_t^X = 0$ at every later date. The multiplicative fiscal innovations in the rules above are held at one. Fiscal smoothing applies to $X_t - D_t^X$ and its lag, so it does not repeat the direct payment.

Taxation. The tax system follows Bayer et al. (2024). The average tax rate τ_t evolves according to

$$\frac{\tau_t}{\bar{\tau}} = \left(\frac{\tau_{t-1}}{\bar{\tau}}\right)^{\rho_\tau} \left(\frac{BFgov_t}{B\bar{F}gov}\right)^{(1-\rho_\tau)\gamma_B^\tau} \left(\frac{Y_t}{Y_{t-1}}\right)^{(1-\rho_\tau)\gamma_Y^\tau}, \tag{11}$$

and tax progressivity follows

$$\frac{\tau_t^P}{\bar{\tau}^P} = \left(\frac{\tau_{t-1}^P}{\bar{\tau}^P}\right)^{\rho_P} \varepsilon_t^P. \tag{12}$$

Aggregate tax revenue is denoted by Tax_t. Tax revenue is raised from labor income and profits. The benchmark tax-rate debt-feedback coefficient is zero; funded adjustment operates through purchases and transfers. Appendix A.1 describes the household-level tax structure in detail.

1.1.2 Monetary Authority

The monetary authority sets the nominal interest rate according to a Taylor-type rule,

$$\frac{R^b_{t+1}}{\bar{R}^b} = \left(\frac{R^b_t}{\bar{R}^b}\right)^{\rho_R} \left(\frac{\pi_t}{\bar{\pi}}\right)^{(1-\rho_R)\theta_\pi} \left(\frac{Y_t}{Y_{t-1}}\right)^{(1-\rho_R)\theta_Y} \epsilon^R_t. \tag{13}$$

I focus on a passive monetary regime in the sense of Leeper (1991), $\theta_\pi < 1$.

1.2 Households and Portfolio Choice

A continuum of households faces idiosyncratic labor-income risk and chooses how much to save in a liquid nominal asset and in illiquid real capital. Workers earn labor income and receive union profits. Entrepreneurs do not work but do receive firm profits.

1.2.1 Income, Assets, and Budget Constraint

The household side follows Bayer et al. (2024). Workers face idiosyncratic productivity risk. Entrepreneurs do not supply labor and receive non-union firm profits. Let b_{it} denote household i's liquid asset holdings and k_{it} denote illiquid real capital. Households can borrow through the liquid asset up to the limit $\underline{B}$, while illiquid capital must remain nonnegative.

Household i's period-t budget constraint is

$$\begin{aligned} c_{it} + b_{it+1} + q_t k_{it+1} = \frac{R_{it}}{\pi_t} b_{it} + (q_t + r_t) k_{it} + y_{it} \\ + \mathbb{I}_{h_{it}\neq 0}(1-\tau_t)\Pi^U_{it} + \mathbb{I}_{h_{it}=0}\tau^L_t (\Pi^E_t)^{1-\bar{\tau}^P} + Transfers_{it}, \end{aligned} \tag{14}$$

subject to

$$k_{it+1} \geq 0, \qquad b_{it+1} \geq \underline{B}. \tag{15}$$

Here, q_t is the price of capital, r_t is the dividend on capital, y_{it} is after-tax labor income, Π_{it}^U denotes union profits received per worker, Π_t^E denotes entrepreneurial profits, and $Transfers_{it}$ denotes government transfers.

Households choose b_{it+1} rather than direct holdings of long-term government debt. The return on the liquid asset is

$$R_{it} = \begin{cases} A_t \dfrac{R_t^b Bgov_t + \pi_t \left[(1-\iota^\Pi) q_t^\Pi + \omega^\Pi \Pi_t^F\right]}{Bgov_t + q_{t-1}^\Pi} & \text{if } b_{it} \geq 0, \\ A_t \dfrac{R_t^b Bgov_t + \pi_t \left[(1-\iota^\Pi) q_t^\Pi + \omega^\Pi \Pi_t^F\right]}{Bgov_t + q_{t-1}^\Pi} + \pi_t \bar{R} & \text{if } b_{it} < 0. \end{cases} \tag{16}$$

Here, A_t is an aggregate liquid-return wedge. The liquid asset return equals the average return on the aggregate liquid portfolio, composed of government debt and tradable claims on firm profits, plus a borrowing wedge for households with negative liquid asset holdings.

1.2.2 Portfolio Adjustment Friction

Only a fraction λ of households can adjust illiquid capital in a given period. Households that receive the adjustment opportunity choose both liquid assets and illiquid capital. Households that do not receive the adjustment opportunity keep illiquid capital fixed and choose only liquid asset holdings. The parameter λ governs how easily households reallocate wealth between liquid nominal assets and illiquid real capital.

1.2.3 Household Problem

Let $\Theta_t(b, k, h)$ denote the joint distribution of liquid asset holdings, capital, and idiosyncratic productivity. Define $V_t^a(b, k, h)$ as the value function for households that can adjust both

assets, and $V_t^n(b,k,h)$ as the value function for households that cannot adjust capital:

$$V_t^a(b,k,h) = \max_{b'_a,k'} u[x(b,b'_a,k,k',h)] + \beta \mathbb{E}_t \mathbb{W}_{t+1}(b'_a,k',h'), \tag{17}$$

$$V_t^n(b,k,h) = \max_{b'_n} u[x(b,b'_n,k,k,h)] + \beta \mathbb{E}_t \mathbb{W}_{t+1}(b'_n,k,h'), \tag{18}$$

$$\mathbb{W}_{t+1}(b',k',h') = \lambda V_{t+1}^a(b',k',h') + (1-\lambda) V_{t+1}^n(b',k',h'). \tag{19}$$

The cross-sectional distribution evolves according to household policy rules and the transition matrix for idiosyncratic productivity:

$$\begin{aligned}\Theta_{t+1}(b',k',h') = \lambda \int_{b'=b^*_{a,t}(b,k,h),\, k'=k^*_t(b,k,h)} \Phi(h,h')\, d\Theta_t(b,k,h) \\ + (1-\lambda) \int_{b'=b^*_{n,t}(b,k,h),\, k'=k} \Phi(h,h')\, d\Theta_t(b,k,h).\end{aligned} \tag{20}$$

Appendix A.1 provides additional details on preferences, labor supply, tax progressivity, and the derivation of household and aggregate income.

1.3 Firms, Banks, and Long-Term Government Debt

1.3.1 Firms

The production side follows a standard New Keynesian structure with monopolistic competition, sticky prices, sticky wages, and investment adjustment costs. Intermediate firms produce using labor and capital,

$$Y_t = Z_t N_t^{1-\alpha} (u_t K_t)^{\alpha}, \tag{21}$$

where Z_t is total factor productivity, N_t is labor input, K_t is the capital stock, and u_t is capital utilization.

Price setting and wage setting imply a standard price Phillips curve and wage Phillips

curve. Capital accumulation is subject to convex adjustment costs. Appendix A.4 provides the firm-side details.

1.3.2 Banks and Long-Term Nominal Debt

Banks intermediate aggregate household liquid asset holdings into long-term nominal government debt, following Bayer et al. (2023). Government bonds have geometric maturity. Each bond pays one unit of nominal coupon per period, and a fraction δ_B matures each period. Let q_t^B denote the nominal bond price. Perfect competition in banking implies

$$R_{t+1}^b q_t^B = \mathbb{E}_t \left[(1 - \delta_B) q_{t+1}^B + 1 \right]. \tag{22}$$

Real bank profits are

$$\Pi_t^B = \frac{Bgov_t}{\pi_t} \left[(1 - \delta_B) \frac{q_t^B}{q_{t-1}^B} + \frac{1}{q_{t-1}^B} - R_t^b \right]. \tag{23}$$

Banks map household liquid asset demand into the pricing of long-term nominal government debt.

1.4 Market Clearing and Equilibrium

Liquid asset market. The liquid asset market clears when

$$B_{t+1} = \mathbb{E}_t \left[\lambda b_{a,t}^* + (1 - \lambda) b_{n,t}^* \right]. \tag{24}$$

Aggregate liquid asset supply consists of long-term government debt and tradable profit claims. With the timing convention in the model, household liquid asset holdings carried into period t equal the government debt carried into period t plus the market value of tradable

profit claims purchased in period $t-1$:

$$B_t = Bgov_t + q^{\Pi}_{t-1}, \tag{25}$$

where q^{Π}_{t-1} is the price of tradable profit claims at the end of period $t-1$. Since the supply of tradable profit claims is normalized to one, q^{Π}_{t-1} is also their aggregate market value.

Tradable profit claims. The price of tradable profit claims satisfies

$$q^{\Pi}_t R^b_{t+1} = \mathbb{E}_t \pi_{t+1} \left[(1-\iota^{\Pi}) q^{\Pi}_{t+1} + \omega^{\Pi} \Pi^F_{t+1} \right]. \tag{26}$$

Here, ι^{Π} is the fraction of outstanding tradable profit claims that matures each period, so $1-\iota^{\Pi}$ is the fraction of claims that remains outstanding into the next period. The parameter ω^{Π} denotes the share of firm profits paid to tradable profit-claim holders.

The original liquid-equity calibration is retained. Its reference parameters are $(\omega^{\text{ref}}_{\Pi}, \iota^{\text{ref}}_{\Pi}) = (0.15, 0.016)$. The maturity mapping sets

$$\iota^{\Pi} = \left(\frac{1}{40} - \frac{1}{800} \right) 0.50 + \frac{1}{800} = 0.013125, \tag{27}$$

$$\omega^{\Pi} = \frac{\iota^{\Pi}}{\iota^{\text{ref}}_{\Pi}} \omega^{\text{ref}}_{\Pi} = 0.123046875. \tag{28}$$

At the baseline steady state, liquid equity accounts for approximately 33.68% of liquid assets; the reference dividend parameter 0.15 is not a portfolio weight.

Capital market. The capital market clears when aggregate household demand for illiquid capital equals the aggregate capital stock. Households that can adjust illiquid assets choose $k^*_t(b,k,h)$, while non-adjusting households keep their current illiquid capital position k_t. Thus,

$$K_{t+1} = \int \left[\lambda k^*_t(b,k,h) + (1-\lambda) k_t \right] d\Theta_t(b,k,h), \tag{29}$$

where $\Theta_t(b, k, h)$ is the joint distribution of liquid asset holdings, illiquid capital, and idiosyncratic productivity at the beginning of period t.

Goods and labor markets. The labor market clears at the equilibrium real wage. The goods market clears through the resource constraint. Market clearing in liquid assets, capital, and goods jointly determines how changes in fiscal backing translate into inflation.

A sequential equilibrium consists of household policy functions, a law of motion for the cross-sectional distribution, sequences of prices and aggregates, and government policies such that households optimize, firms and banks optimize, government budget constraints and policy rules hold, and all markets clear.

2 Calibration

I calibrate the model at quarterly frequency. The steady state is chosen to match the same broad long-run objects as Bayer et al. (2024), with particular attention to household portfolio structure, the prevalence of borrowers, long-term nominal debt, and the fiscal side. The targeted moments include earnings persistence and dispersion, worker–entrepreneur transitions, the liquid–illiquid composition of household wealth, factor shares, government size, and tax progressivity.

2.1 Steady-State Calibration

The household side is disciplined by a small set of preference, income-process, and portfolio parameters. I set

$$(\xi, \gamma, \beta) = (4,\ 2,\ 0.98255),$$

which pin down precautionary saving, labor supply, and the capital–output ratio. The idiosyncratic income process is governed by $(\rho_h, \sigma_h) = (0.98,\ 0.12)$, and worker–entrepreneur transitions are set to $\iota = 1/16$ and $\zeta = 1/4500$, so that the model matches earnings persis-

tence, earnings dispersion, and a thin entrepreneurial tail.

The portfolio structure is disciplined by three key parameters. The probability of illiquid asset adjustment is $\lambda = 0.03$, which controls how often households can rebalance between liquid nominal assets and illiquid capital. Long-term government debt has geometric maturity with $\delta_B = 1/5$, which implies an average duration of five quarters. Tradable profit claims retain reference parameters $(\omega_{\Pi}^{\text{ref}}, \iota_{\Pi}^{\text{ref}}) = (0.15, 0.016)$ and the maturity mapping described above. The resulting liquid-to-capital ratio is $B/K = 0.2488$, including government liabilities and liquid equity. Together, these parameters pin down the liquid–illiquid composition of household portfolios and the prevalence of borrowers.

On the production side, I set the capital share to $\alpha = 0.318$, consistent with a labor share of about 0.68, and quarterly depreciation to $\delta_0 = 0.0215$. Gross price and wage markups are set to $\mu^Y = \mu^W = 1.1$. On the fiscal side, tax parameters are $(\tau^L, \tau^P) = (0.645, 0.1022)$, which imply a government consumption share near 15 percent and a realistic degree of tax progressivity. I normalize steady-state gross inflation and the gross policy rate to $\pi = \mathrm{R}^{\mathrm{b}} = 1$. The borrower wedge is $\bar{R} = (1.20)^{1/4} - 1 \approx 0.0466$.

Table 1 summarizes the main calibrated parameters used in the quantitative exercises. Appendix Table 3 reports the full steady-state calibration, targets, and data sources.

2.2 Policy Rules for Funded and Unfunded Deficits

Funded and unfunded deficits are implemented through different debt-feedback coefficients in the fiscal rules. Since the rules are written for government purchases and transfers, a negative debt-feedback coefficient means that higher debt reduces purchases or transfers. This raises primary surpluses and therefore funds the initial deficit. A zero debt-feedback coefficient means that the spending component does not adjust directly to higher debt.

In the baseline calibration, funded government purchases and transfers load on negative debt-feedback coefficients, $\gamma_{BGF} = \gamma_{BTF} = -0.50$, while both unfunded government purchases and transfers have zero debt-feedback coefficients, $\gamma_{BGU} = \gamma_{BTU} = 0$. The other

Table 1: Key calibrated parameters

Category	Symbol	Value	Role / target
Preferences	ξ	4.0	Precautionary saving
	γ	2.0	Labor supply elasticity
	β	0.98255	K/Y
Income risk	ρ_h	0.98	Earnings persistence
	σ_h	0.12	Earnings dispersion
	ι, ζ	1/16, 1/4500	Worker–entrepreneur flows
Portfolio structure	λ	0.03	Illiquid asset adjustment frequency
	δ_B	0.20	Debt maturity
	$\omega_\Pi^{\text{ref}}, \iota_\Pi^{\text{ref}}$	0.15, 0.016	Tradable profit-claim supply
Technology	α	0.318	Labor share
	δ_0	0.0215	Depreciation
	μ^Y, μ^W	1.1, 1.1	Price and wage markups
Fiscal side	τ^L, τ^P	0.645, 0.1022	Government size and progressivity
	$\bar{R}$	0.0466	Borrowing wedge

fiscal-rule coefficients are held fixed across the four deficit experiments: all government-purchase and transfer rules have persistence $\rho = 0.90$ and output response $\gamma_Y = 0.50$. Thus, the main difference between funded and unfunded experiments comes from the debt-feedback coefficient. Table 2 reports the baseline fiscal-rule coefficients, the monetary policy rule, and the shock processes used in the quantitative exercises.

The benchmark experiments use a one-time additive intervention at date zero, with no subsequent direct payments. The fiscal-rule persistence remains 0.90; it governs endogenous adjustment rather than the duration of the direct payment. I hold the total initial fiscal cost fixed across instruments and recipient groups. Background AR(1) shocks remain inactive. Appendix Table 5 reports the full set of policy-rule and background shock-process parameters.

2.3 Solution Method

I solve the model following the Reiter-style approach in Bayer et al. (2024). I first compute the stationary equilibrium by iterating on household policy functions and the invariant

Table 2: Baseline policy-rule parameters

Rule	**Parameters**	**Values**
Funded government purchases	$(\rho_{GF}, \gamma_{YGF}, \gamma_{BGF})$	(0.90, 0.50, −0.50)
Unfunded government purchases	$(\rho_{GU}, \gamma_{YGU}, \gamma_{BGU})$	(0.90, 0.50, 0.00)
Funded transfers	$(\rho_{TF}, \gamma_{YTF}, \gamma_{BTF})$	(0.90, 0.50, −0.50)
Unfunded transfers	$(\rho_{TU}, \gamma_{YTU}, \gamma_{BTU})$	(0.90, 0.50, 0.00)
Monetary policy	$(\rho_R, \theta_\pi, \theta_Y)$	(0.70, 0.50, 0.125)

cross-sectional distribution. I then reduce the high-dimensional distributional state, linearize the equilibrium system around the stationary equilibrium, and solve the resulting linear rational-expectations system. All reported general-equilibrium experiments use this reduced first-order solution. Household budget and distributional observables are differentiated consistently around the same stationary equilibrium. Appendix C reports the numerical algorithm, state reduction procedure, and diagnostic checks.

3 Results

In this section, I examine the responses of aggregate variables to funded and unfunded fiscal shocks to transfers and government purchases. I also examine the distributional consequences of transfer shocks. Unless noted otherwise, $\lambda = 0.03$, both unfunded debt-feedback coefficients are zero, and liquid equity is retained. Percentage impulse responses use a one-time direct payment of 5 percent of each economy's quarterly steady-state output. This scale approximates the first round of U.S. Economic Impact Payments relative to pre-pandemic quarterly GDP.* These are linearly scaled first-order responses around the stationary equilibrium; nonlinear accuracy at this payment size has not been established. Every dynamic figure includes dates zero through forty.

*The U.S. Treasury reports approximately $271.4 billion in first-round payments. Dividing by quarterly GDP of approximately $5.43 trillion, based on the BEA's 2019:Q4 annualized current-dollar GDP of $21.73 trillion, gives about 5 percent. Sources: U.S. Treasury and BEA. This is a scale benchmark, not a replication of payment eligibility or a classification of the actual policy's fiscal backing.

3.1 Aggregate Effects on Output, Consumption, Investment, and Inflation

This section reports impulse response functions for four fiscal deficit shocks. The shocks differ along two dimensions: whether the deficit comes from government purchases or transfers, and whether the deficit is funded or unfunded.

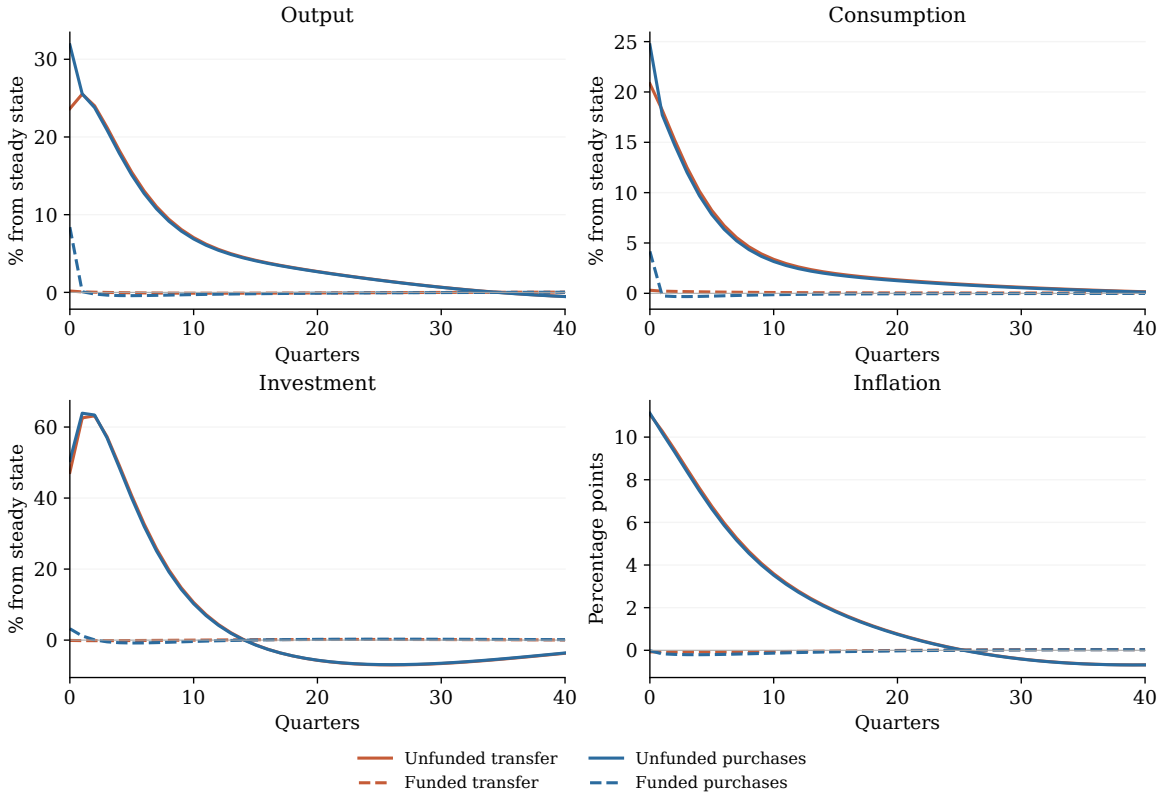
Figure 1 reports the impulse responses of output, consumption, investment, and inflation to fiscal shocks. Two results stand out. First, unfunded fiscal shocks generate much larger initial responses than their funded counterparts. Second, government purchases and transfers differ in their response levels, but the unfunded-minus-funded increment in impact consumption is nearly the same for both instruments. Impact consumption per unit of direct fiscal cost is 0.0385 for funded transfers and 0.5256 for funded purchases; the corresponding unfunded responses are 2.627 and 3.114. The backing increment is approximately 2.589 for either instrument.

The contrast between funded and unfunded shocks comes from fiscal backing. When the deficit is funded, higher expected future primary surpluses sustain household demand for nominal government bonds. The shift away from nominal bonds is therefore limited, goods demand rises by less, and output, consumption, investment, and inflation move only modestly. When the deficit is unfunded, weak backing reduces household demand for nominal government bonds and shifts demand toward goods. As a result, real activity increases more strongly and the price level must increase by more as well.

This same mechanism helps explain the similar backing increment across fiscal instruments. Once fiscal backing is weak, the common adjustment in liquid asset demand becomes quantitatively important. Differences between purchases and transfers nevertheless remain visible: purchases enter goods demand directly, whereas transfers initially enter household budgets.

By contrast, when the deficit is funded, the source of the deficit matters more. A funded transfer raises household disposable income directly, so its effect depends on who receives

Figure 1: Impulse Response Functions (IRFs) to fiscal deficit shocks



Note: First-order impulse responses to a one-time direct intervention equal to 5 percent of quarterly steady-state output, using a 320-quarter first-order solution. Inflation is in percentage points; other responses are percent deviations from the no-intervention path. Solid lines represent unfunded shocks, and dashed lines represent funded shocks. Red lines indicate transfers, and blue lines indicate government consumption. Transfer shocks are distributed evenly across households.

the transfer and how strongly those households spend out of additional income. Transfers work through households' marginal propensities to consume: a larger share of the transfer is spent when it reaches households with high MPCs, while more of it is saved when it reaches households with low MPCs. A funded government-purchases shock operates differently. It raises demand because the government directly purchases goods, rather than because households receive additional disposable income. Since the deficit is funded, however, the increase in government purchases is backed by higher expected future primary surpluses. Households do not treat the fiscal expansion as an unbacked increase in nominal wealth. Future fiscal adjustment limits the rise in private spending and can crowd out consumption and investment. At the same time, sufficient fiscal backing prevents the large portfolio reallocation away from nominal government bonds that occurs under unfunded deficits. The direct increase in government demand is partly offset by weaker private demand. In this experiment, funded purchases raise impact aggregate consumption more than funded transfers; the direct-transfer MPC alone therefore does not rank the general-equilibrium effects of the two instruments.

3.1.1 Investment and the Labor Market

The portfolio response also has a production counterpart. Figure 2 reports investment, capital, labor input, and the real wage paid by firms. Under unfunded transfers, investment, labor input, and the firm-paid real wage rise initially. Capital peaks around quarter eight. Investment subsequently falls below baseline around quarter fifteen, and capital does so around quarter twenty-three. The initial expansion is therefore followed by a reversal within the forty-quarter window. The wage received by workers differs from the firm-paid wage because of the wage markup.

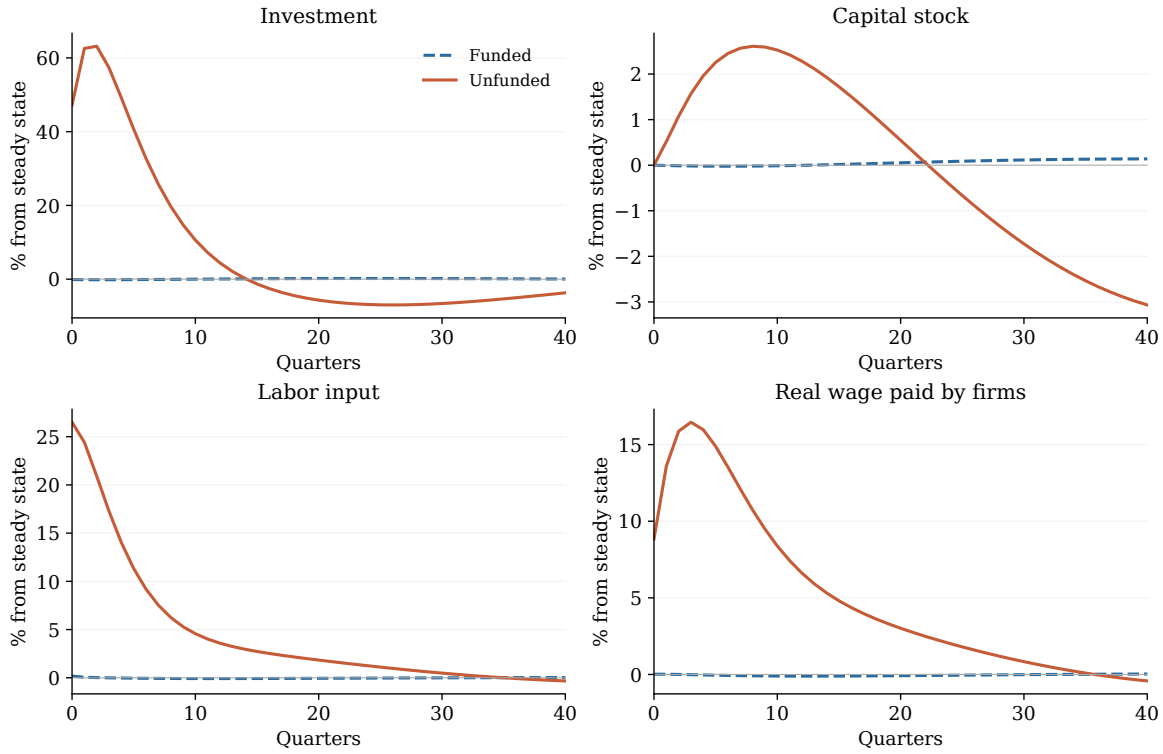


Figure 2: Investment, capital, and labor-market responses to transfers

Note: Uniform one-time transfers; a 320-quarter first-order solution. Responses are percent deviations from the no-intervention path, normalized to a payment of 5 percent of quarterly output. Dashed blue lines are funded; solid red lines are unfunded.

3.1.2 Where Does a Transfer of 100 Go?

To connect these aggregate responses to household choices, I reconstruct the household budget and normalize the direct transfer to 100. Define liquid net saving as $b_{t+1} - b_t$ and capital net purchases as $q_t(k_{t+1} - k_t)$. Let other resources collect net earnings, net liquid returns, capital income, and endogenous transfers beyond the direct payment. Changes in these quantities satisfy

$$\Delta c + \Delta\text{liquid net saving} + \Delta\text{capital net purchases} - \Delta\text{other resources} = 100. \tag{30}$$

Under funded policy, consumption rises by 3.85, liquid net saving rises by 101.26, capital net purchases fall by 1.47, and other resources rise by 3.65. Under unfunded policy, consumption rises by 262.72, liquid net saving falls by 365.89, capital net purchases rise by 105.29, and other resources fall by 97.88. Figure 3 places positive and negative contributions on opposite sides of zero. Consumption above 100 is accompanied by a substantial reduction in liquid net saving. It is not financed by an increase in net other resources.

These are general-equilibrium changes relative to the no-intervention path, not fixed-price household MPCs. A negative change in net saving or net purchases need not imply a negative level. Nor does the budget identity alone establish which equilibrium channel causes the consumption response. Appendix D.1 separates households that can and cannot adjust capital.

3.2 Distributional Effects: Mean Consumption Across the Liquid-Asset Distribution and Consumption Inequality

Figure 4 reports the impulse responses of mean consumption in the bottom 10 percent, the 45th–55th percentiles, and the top 10 percent of the liquid-asset distribution, together with the consumption Gini. The liquid-asset distribution ranks households by their current holdings of liquid assets. Groups are re-ranked at each date, so group means include changes

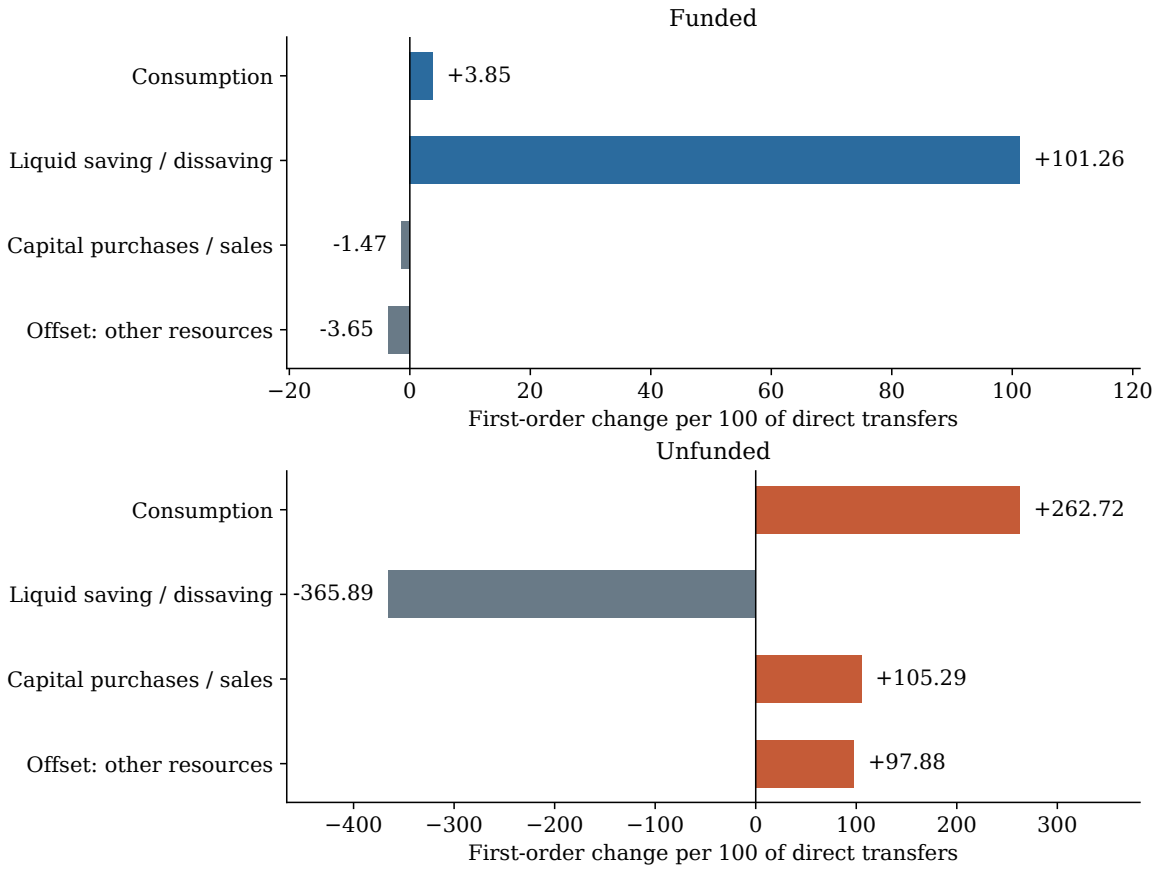


Figure 3: Household budget changes per 100 of direct transfers

Note: First-order impact responses for all households, $\lambda = 0.03$, using a 320-quarter solution. The other-resources contribution enters with a minus sign so that the four signed contributions sum to 100. Numbers describe changes, including changes in endogenous income and transfers.

in group composition. Households at the bottom of this distribution hold little liquid wealth or are borrowing constrained, while households at the top hold large positive liquid nominal positions. Transfer shocks are lump-sum and distributed evenly across households.

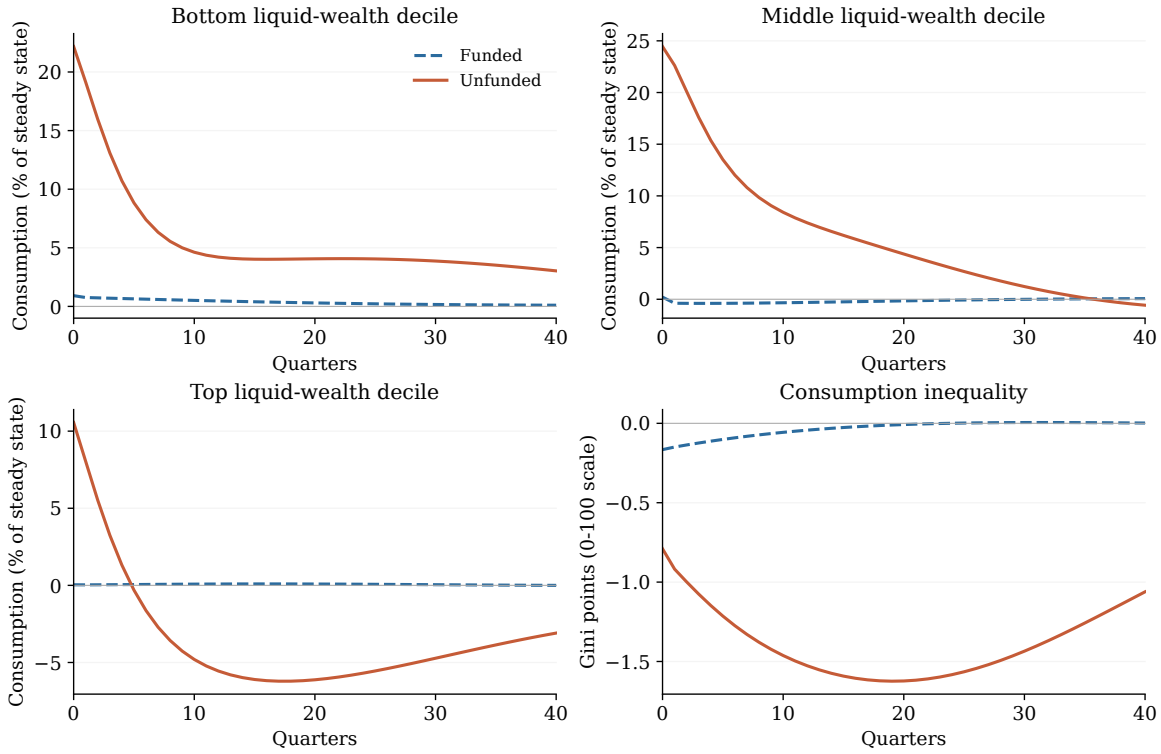
Unfunded transfers initially raise mean consumption strongly in the bottom 10 percent and in the 45th–55th percentiles. Mean consumption at the top also rises on impact, but falls below its no-intervention path from quarter five onward through quarter forty. Funded transfers raise impact consumption in all three groups, with a larger response among households with low liquid wealth. The consumption Gini initially falls under both policies and remains below baseline through quarter forty under unfunded policy; the funded response turns slightly positive from quarter twenty-four.

When deficits are unfunded, mean consumption initially rises strongly in the bottom 10 percent and in the 45th–55th percentiles as the transfer and the general-equilibrium adjustment change spending and portfolio choices. For households at the top of the liquid-asset distribution, the dominant force is different. Since they hold large positive nominal positions, inflation reduces the real value of those assets and lowers their purchasing power. Mean consumption therefore falls for the top 10 percent after the initial positive response. Consumption inequality declines as consumption rises more strongly for households with low and middle liquid wealth; the later decline at the top reinforces this pattern.

When deficits are funded, the source of the deficit matters more. Funded transfers raise disposable income directly and support households with low liquid assets more strongly. Since transfers are distributed evenly across households, the same dollar transfer represents a larger effective increase in resources for households with little liquid wealth. These households also tend to have higher MPCs, so their consumption responds more strongly. As a result, funded transfers initially reduce consumption inequality.

These transfer results combine direct receipt, changes in labor and profit income, asset returns, and portfolio adjustment. The comparison across current liquid-wealth groups describes distributional incidence, including changes in group composition; it neither tracks

Figure 4: Transfer responses: mean consumption across the liquid-asset distribution and consumption inequality



Note: Uniform one-time transfers; a 320-quarter first-order solution. Groups are defined by current liquid wealth and re-ranked each period. Solid red lines show unfunded transfers; dashed blue lines show funded transfers. Consumption is in percent deviations for a payment of 5 percent of quarterly output, and the Gini is in points on a 0–100 scale.

fixed household cohorts nor identifies the separate contribution of each channel.

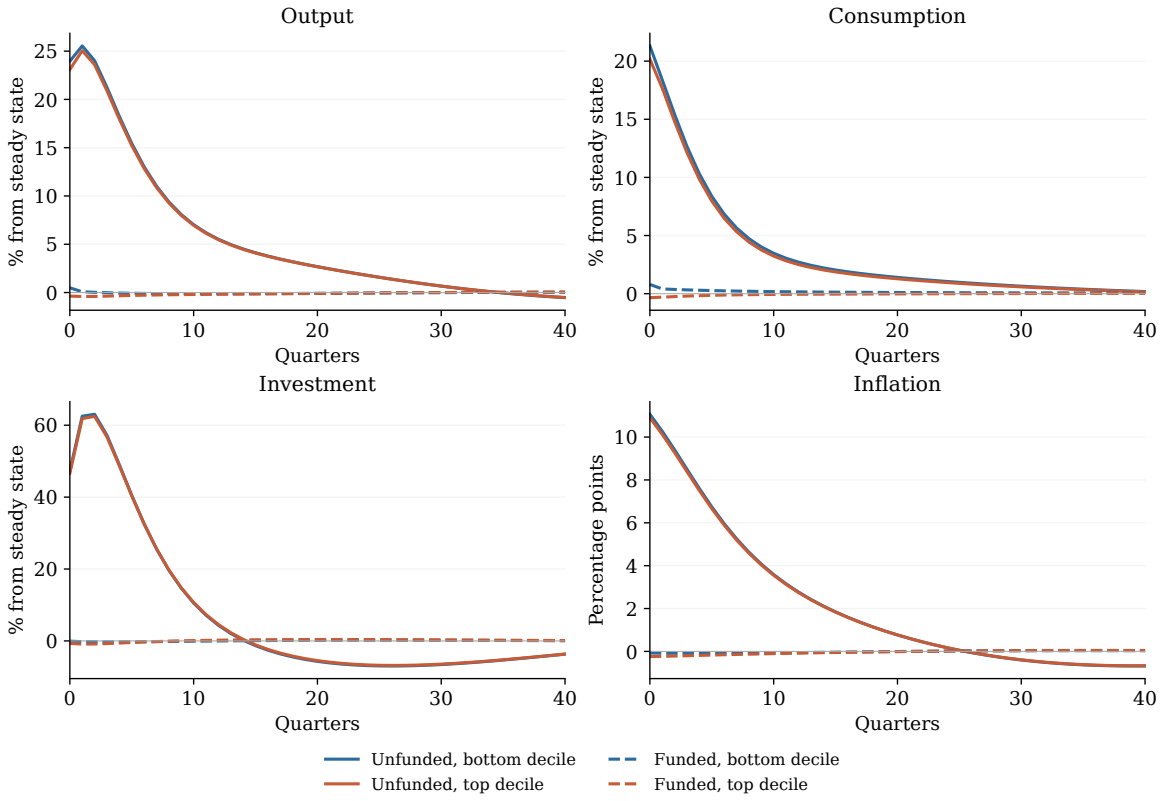
3.3 Impact of Transfer Targeting and Fiscal Backing

Figure 5 reports the impulse responses of output, consumption, investment, and inflation to transfer shocks targeted to either the bottom 10 percent or the top 10 percent of the liquid-asset distribution. Total direct fiscal cost is held fixed. Each decile recipient receives ten times the uniform per-household payment. Two results stand out. First, when transfers are funded, targeting matters a great deal for consumption and investment. Transfers to the bottom 10 percent generate a much larger rise in consumption than transfers to the top 10 percent. Unfunded transfers to either group generate a much larger response, while the difference across recipient groups remains visible.

When transfers are funded, the main channel is the direct spending response of the recipients. Households at the bottom of the liquid-asset distribution have higher marginal propensities to consume, whereas households at the top have lower marginal propensities to consume. Transfers to the bottom 10 percent therefore raise consumption much more strongly, whereas transfers to the top 10 percent produce an impact decline in aggregate consumption. The impact responses per unit of total payment are 0.1001 and -0.0431, respectively. These aggregate responses include nonrecipients and are not recipient MPCs.

When transfers are unfunded, fiscal backing adds a second force to this MPC heterogeneity. Households become less willing to hold nominal government bonds at the pre-shock price level, demand shifts toward goods, and inflation rises. That portfolio-reallocation channel pushes up output and inflation even when the transfer is targeted to the top 10 percent. As a result, targeting matters less for the aggregate responses to unfunded transfers than it does when transfers are funded. At the same time, the consumption response remains stronger when the transfer goes to the bottom 10 percent: impact consumption per unit of payment is 2.689, compared with 2.546 for the top decile. The increment relative to funded policy is approximately 2.589 in both cases.

Figure 5: Aggregate responses to targeted funded and unfunded transfer shocks



Note: One-time targeted transfers with equal total fiscal cost, normalized to 5 percent of quarterly output. The first-order solution horizon is 320 quarters. Inflation is in percentage points; other responses are percent deviations. Solid lines represent unfunded shocks, and dashed lines represent funded shocks. Blue lines indicate transfers to the bottom 10 percent of the liquid-asset distribution, and red lines indicate transfers to the top 10 percent. Total direct fiscal cost is held fixed. Each decile recipient receives ten times the uniform per-household payment.

This comparison shows that targeting and fiscal backing affect transmission in different ways. Targeting mainly changes the direct spending response of the recipients. Fiscal backing mainly changes whether the transfer also induces a broader shift away from nominal government bonds and into goods. Once the transfer is unfunded, that second channel becomes strong enough that transfers to low- and high-liquid-asset households generate similar movements in output and inflation.

3.4 Role of Portfolio Adjustment Opportunities

This section studies how the funded–unfunded difference depends on households' ability to rebalance between liquid nominal assets and illiquid capital. The key parameter is the portfolio adjustment probability λ, which governs the fraction of households that can adjust illiquid asset holdings in a given period. A higher value of λ means that households have more frequent access to the illiquid capital market and can move wealth between liquid nominal assets and illiquid real assets more easily.

Figure 6 compares impulse responses for two values of λ: the baseline value, $\lambda = 0.03$, and a lower-adjustment case, $\lambda = 0.005$. The comparison shows that portfolio adjustment opportunities matter for both funded and unfunded transfers. At $\lambda = 0.005$, impact consumption per unit of payment is 0.0637 under funded policy and 2.335 under unfunded policy. At $\lambda = 0.03$, it is 0.0385 and 2.627. More frequent adjustment therefore lowers funded consumption and raises unfunded consumption in the reduced first-order solution; the unfunded-minus-funded difference widens from 2.272 to 2.589.

The reason is that changing λ affects two margins at once. First, more frequent portfolio adjustment improves households' ability to self-insure against idiosyncratic income risk. When households can access illiquid capital more easily, they rely less on current disposable income to smooth consumption. This dampens the effects of funded deficits, because funded deficits are backed by higher expected future primary surpluses and do not generate a large revaluation of nominal government debt. In this case, easier access to illiquid assets mainly

strengthens households' ability to smooth consumption over time, consistent with a smaller funded impact consumption response.

Second, more frequent portfolio adjustment changes the response to unfunded deficits. When fiscal backing is weak, households can reduce their liquid positions and increase their purchases of illiquid capital. A higher λ makes this reallocation available to a larger fraction of households. In the reduced first-order experiment, unfunded impact consumption is larger in the economy with more frequent adjustment. This pattern is consistent with a stronger portfolio response, although the comparison alone does not identify that channel separately.

A higher λ also changes steady-state portfolio composition. When households can adjust illiquid assets more frequently, they need to hold fewer liquid nominal assets for precautionary purposes and can hold more wealth in illiquid capital. The steady-state liquid-to-illiquid asset ratio therefore falls. This portfolio composition matters for the transmission of fiscal shocks. In the higher-adjustment economy, households enter the shock with a smaller reliance on liquid nominal assets for self-insurance, but they also have a stronger ability to rebalance toward real capital when fiscal backing weakens. The observed comparison combines adjustment opportunities with the associated stationary portfolios. It does not hold the initial wealth distribution fixed, and it does not identify either effect separately.

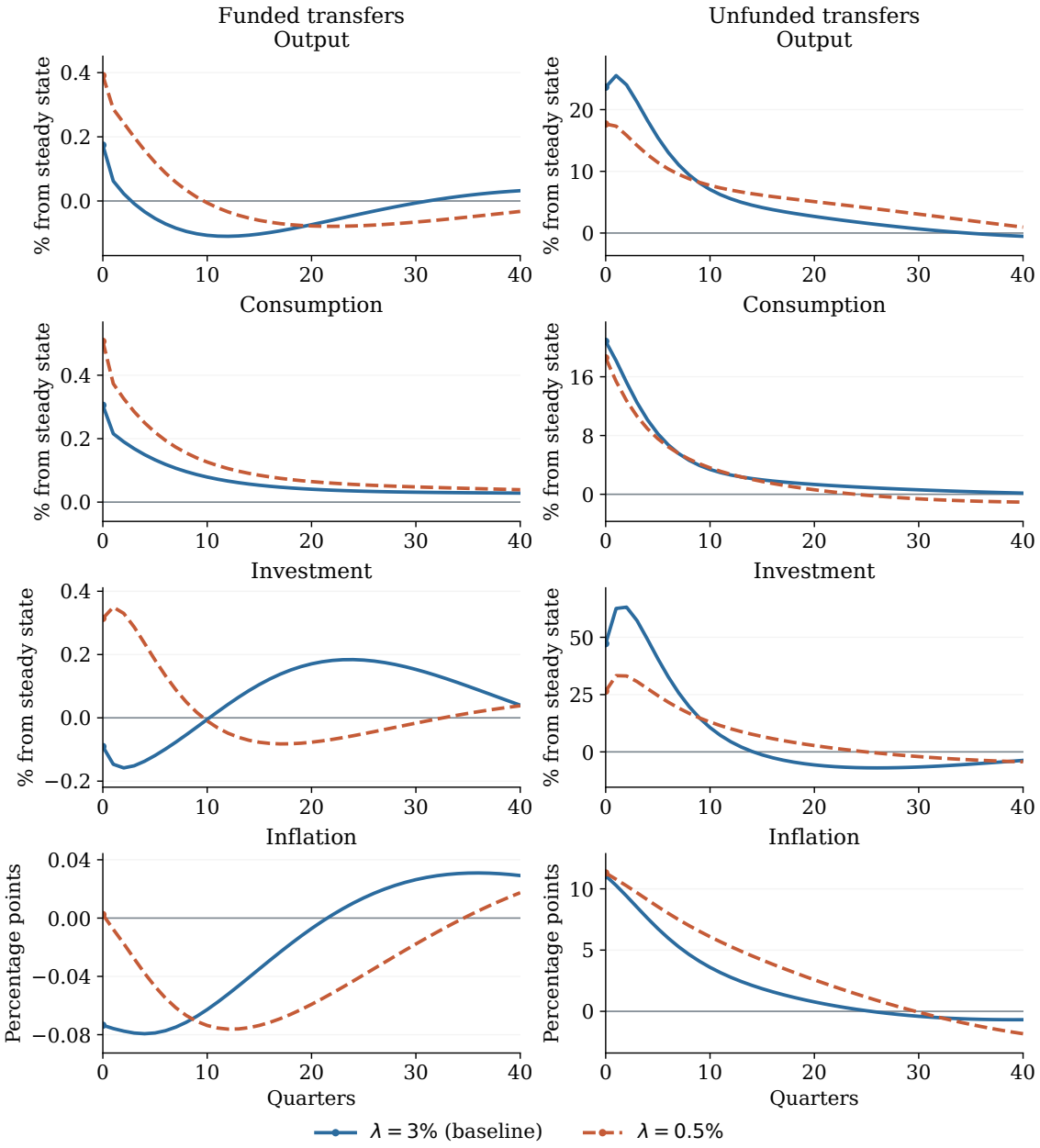


Figure 6: Impulse responses under different portfolio adjustment probabilities. Funded transfers are in the left column and unfunded transfers in the right. Each panel compares $\lambda = 0.03$ (solid blue) with $\lambda = 0.005$ (dashed orange), using separate vertical scales across regimes. Rows show output, consumption, investment, and inflation; horizontal lines mark zero. Both first-order solutions use 320 quarters and a one-time payment of 5 percent of each economy's quarterly steady-state output. Inflation is in percentage points; other responses are percent deviations from steady state.

These results qualify the role of adjustment frequency in the paper's mechanism. The inflationary effects of fiscal deficits depend not only on household heterogeneity, but also on the

assets households can use to respond to fiscal news. When households can reallocate wealth across liquid nominal bonds and illiquid real capital, weak fiscal backing triggers a portfolio response that is absent or muted in environments with only liquid nominal assets. The fiscal theory mechanism therefore remains useful for understanding inflation in heterogeneous-agent economies, because portfolio choice determines how strongly households can move away from nominal government liabilities when deficits are not backed.

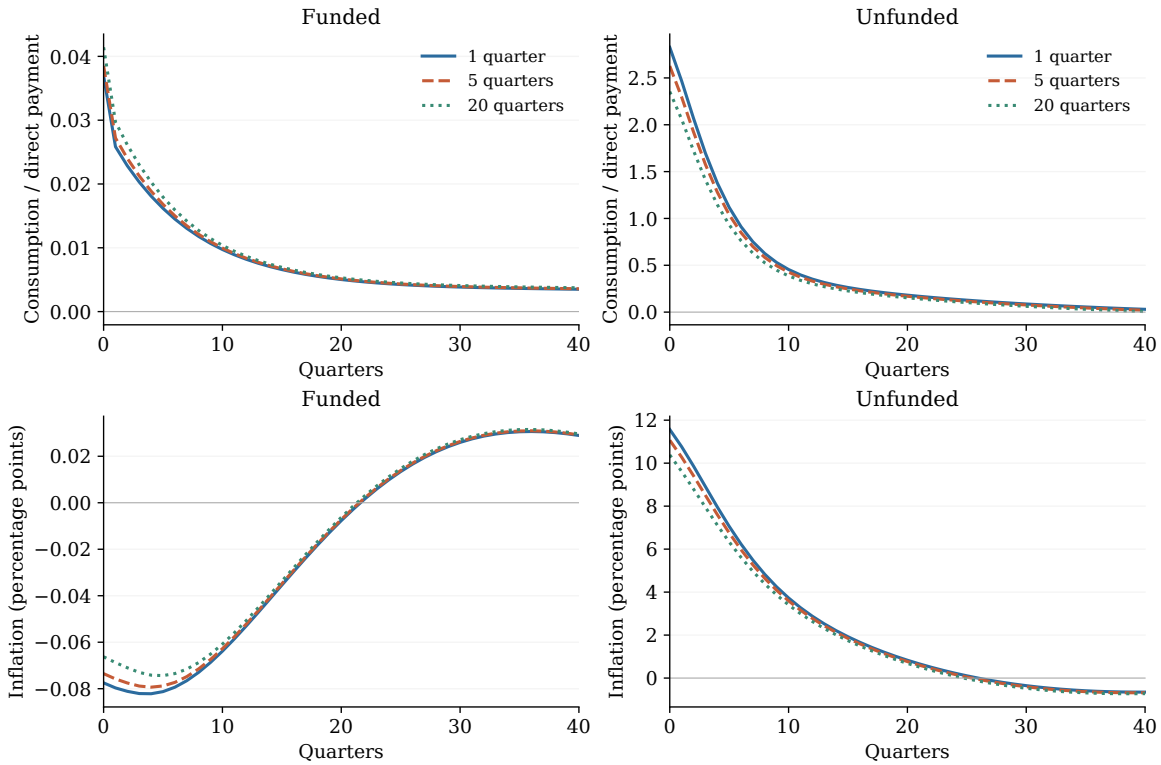
3.5 Role of Debt Maturity

The maturity structure of government debt matters because it affects the valuation of outstanding nominal liabilities. Since the paper's mechanism operates through changes in the real value of nominal government debt and households' portfolio reallocation away from that debt, debt maturity could in principle change the strength of the response to unfunded fiscal shocks. Longer-maturity debt makes a larger share of government liabilities sensitive to the expected path of inflation and nominal interest rates, while short-term debt is repriced more quickly.

Figure 7 reports the responses to funded and unfunded transfers for the baseline value $\delta_B = 0.2$ and two alternative values, $\delta_B = 0.05$ and $\delta_B = 1$, where δ_B is the fraction of long-term debt that matures each period. Lower values of δ_B correspond to longer maturity. When $\delta_B = 1$, all debt matures each period, so the government issues only one-period debt. Debt maturity has a visible but quantitatively limited effect on the responses to unfunded transfers. A longer debt maturity structure lowers unfunded impact consumption per unit of payment from 2.827 at $\delta_B = 1$ to 2.627 at 0.2 and 2.354 at 0.05. These differences do not change the paper's main quantitative results: unfunded fiscal shocks remain substantially more expansionary and inflationary than funded shocks, and the assumed maturity structure of government debt is not the main driver of the results.[†]

[†]For funded transfers, the corresponding impact consumption responses are 0.0369, 0.0385, and 0.0414. These comparisons use the same reduced first-order method as the main experiments; Appendix C.10 reports numerical checks.

Figure 7: Role of debt maturity: funded and unfunded transfers



Note: First-order consumption and inflation responses to a one-time transfer, for alternative values of δ_B, holding the stationary market value of debt fixed. Consumption is divided by the initial payment. Inflation is in percentage points for a payment of 5 percent of quarterly output. Vertical scales differ across regimes.

4 Conclusion

In this paper, I study under what circumstances fiscal deficits become inflationary in a heterogeneous-agent New Keynesian economy with long-term nominal debt and endogenous household portfolio choice. The key distinction is whether deficits are backed by higher expected future primary surpluses. When backing is weak, households reduce their demand for nominal government bonds and rebalance toward goods and real assets. This portfolio response raises inflation and real activity by reducing the real value of outstanding nominal liabilities and shifting demand away from government debt.

The central result is that household portfolio choice changes how fiscal backing is transmitted. In an environment where households hold both liquid nominal government liabilities and illiquid real capital, weak fiscal backing does not only lower the real value of nominal debt. It also induces households to rebalance away from nominal liabilities and toward real assets. The unfunded transfer generates a large consumption response alongside lower liquid net saving and higher capital net purchases.

This mechanism helps explain why unfunded deficits generate much larger aggregate responses than funded deficits. The increase in impact consumption associated with unfunded policy is similar across purchases and transfers, although their response levels differ. Once fiscal backing is weak, the common portfolio-reallocation channel becomes quantitatively important. When deficits are funded, by contrast, expected future primary surpluses limit the revaluation of nominal debt, portfolio reallocation is muted, and the fiscal instrument matters more. Funded transfers provide direct support to recipients, whereas funded purchases enter goods demand directly; in this experiment, funded purchases raise impact aggregate consumption more.

Fiscal backing also shapes distributional outcomes. Unfunded transfers reduce the real value of nominal asset positions and lower consumption inequality. Consumption rises among households with low and middle liquid wealth, while consumption at the top falls after its initial increase. Funded deficits do not generate the same revaluation effect, so their distributional consequences depend more on the fiscal instrument and the allocation of transfers.

The portfolio margin is central to these results. When households can adjust illiquid asset positions more frequently, funded impact consumption is smaller and unfunded impact consumption is larger in the reduced first-order comparison considered here. Easier access to illiquid capital allows households to smooth consumption more effectively when deficits are funded, and also changes the reallocation of wealth when fiscal backing weakens. The net effect must be determined quantitatively. Thus, the fiscal theory mechanism remains useful for understanding inflation because portfolio choice determines how strongly households can

move away from nominal government liabilities when deficits are not backed.

Declaration of Generative AI and AI-Assisted Technologies in the Manuscript Preparation Process

During the preparation of this work, the author used ChatGPT, an AI-assisted writing tool developed by OpenAI, to assist with language editing and improving clarity of exposition. After using this tool, the author reviewed and edited the content as needed and takes full responsibility for the content of the manuscript.

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A Additional Model Details

This appendix describes the preference, income, tax, production, and aggregation blocks that support the model in the main text.

A.1 Preferences, Labor Supply, and Tax Progressivity

Households have Greenwood–Hercowitz–Huffman (GHH) preferences. Household i chooses consumption c_{it} and labor supply n_{it} to maximize

$$\mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t u(c_{it} - G(h_{it}, n_{it})) \,, \tag{31}$$

where β is the discount factor, h_{it} is idiosyncratic labor productivity, and $G(h_{it}, n_{it})$ is the disutility of labor. Period utility is

$$u(x_{it}) = \frac{x_{it}^{1-\xi} - 1}{1-\xi}, \tag{32}$$

with

$$x_{it} = c_{it} - G(h_{it}, n_{it}), \tag{33}$$

where ξ is the risk-aversion parameter.

Following Benabou (2002) and Heathcote et al. (2017), after-tax labor income is

$$y_{it} = \tau_t^L \left(w_t h_{it} n_{it}\right)^{1-\tau_t^P} , \tag{34}$$

where w_t is the aggregate real wage, τ_t^L is the after-tax income coefficient, and τ_t^P is tax progressivity. A higher value of τ_t^P implies a more progressive tax schedule.

The marginal disutility of labor is

$$\frac{\partial G(h_{it}, n_{it})}{\partial n_{it}} = (1-\tau_t^P)\tau_t^L (w_t h_{it})^{1-\tau_t^P} n_{it}^{-\tau_t^P} = (1-\tau_t^P)\frac{y_{it}}{n_{it}}. \tag{35}$$

Assume that the disutility function has constant elasticity with respect to labor supply:

$$\frac{\partial G(h_{it}, n_{it})}{\partial n_{it}} = (1+\gamma)\frac{G(h_{it}, n_{it})}{n_{it}}, \tag{36}$$

with $\gamma > 0$. Then the composite consumption-labor term can be written as

$$x_{it} = c_{it} - \frac{1-\tau_t^P}{1+\gamma} y_{it}. \tag{37}$$

Thus, household decisions can be expressed in terms of the transformed object x_{it} and after-tax income y_{it}.

For the quantitative solution, I follow Bayer et al. (2024) and keep tax progressivity fixed at its steady-state value in the labor-supply block, $\tau_t^P \equiv \bar{\tau}^P$. I use

$$G(h_{it}, n_{it}) = h_{it}^{1-\bar{\tau}^P} \frac{n_{it}^{1+\gamma}}{1+\gamma}. \tag{38}$$

Under this specification, the labor-supply first-order condition implies that individual productivity does not affect optimal hours. All employed households choose the same labor supply, denoted by N_t.

Given the normalization $\int h_{it}\, di \equiv 1$, total effective labor input is

$$\int h_{it} n_{it}\, di = N_t. \tag{39}$$

Substituting the common labor choice into after-tax labor income gives

$$y_{it} = \tau_t^L (w_t h_{it} n_{it})^{1-\bar{\tau}^P} \tag{40}$$

$$= (\tau_t^L)^{\frac{1+\gamma}{\gamma+\bar{\tau}^P}} (1-\bar{\tau}^P)^{\frac{1-\bar{\tau}^P}{\gamma+\bar{\tau}^P}} w_t^{\frac{1+\gamma}{\gamma+\bar{\tau}^P}(1-\bar{\tau}^P)} h_{it}^{1-\bar{\tau}^P}. \tag{41}$$

Income heterogeneity therefore comes from heterogeneity in labor productivity.

A.2 Household Income, Occupation, and Profit Distribution

The household income process follows Bayer et al. (2024). I distinguish between workers and entrepreneurs. Workers face idiosyncratic labor-income risk. Entrepreneurs correspond to the zero-labor state and receive non-union firm profits.

Let $\tilde{h}_{it}$ denote pre-normalized productivity. Productivity evolves as

$$\tilde{h}_{it} = \begin{cases} \exp(\rho_h \log \tilde{h}_{it-1} + \epsilon^h_{it}) & \text{with probability } 1-\zeta \text{ if } h_{it-1} \neq 0, \\ 1 & \text{with probability } \iota \text{ if } h_{it-1} = 0, \\ 0 & \text{otherwise,} \end{cases} \tag{42}$$

where

$$h_{it} = \frac{\tilde{h}_{it}}{\int \tilde{h}_{it}\, di}. \tag{43}$$

Households with $h_{it} \neq 0$ are workers, while households with $h_{it} = 0$ are entrepreneurs. Worker productivity follows a log-AR(1) process with persistence ρ_h and innovation $\epsilon^h_{it} \sim \mathcal{N}(0, \bar{\sigma}^2_h)$. The parameter ζ governs transitions from worker to entrepreneur, and ι governs transitions from entrepreneur to worker.

Workers receive labor income and a lump-sum share of union profits. Entrepreneurs receive firm profits other than union profits. They can liquidate only a fraction ω^Π of firm profits through tradable profit claims with stochastic maturity. A fraction ι^Π of existing claims matures each period. Let q^Π_t denote the price of profit-claim paper. Entrepreneurial profit income is

$$\Pi^E_t = \frac{(1-\omega^\Pi)\Pi^F_t + \iota^\Pi q^\Pi_t + \Pi^B_t}{p_E}, \tag{44}$$

where Π^F_t denotes firm profits, Π^B_t denotes bank valuation profits, and p_E is the entrepreneur population share. Thus Π^E_t is income per entrepreneur. In the household budget, union profits are likewise distributed per worker, $\Pi^U_{it} = \Pi^U_t/(1-p_E)$.

Using

$$c_{it} = x_{it} + \frac{1-\bar{\tau}^P}{1+\gamma} y_{it}, \tag{45}$$

the household budget constraint can be written as

$$\begin{aligned} x_{it} + b_{it+1} + q_t k_{it+1} = \frac{R_{it}}{\pi_t} b_{it} + (q_t + r_t) k_{it} + \frac{\bar{\tau}^P + \gamma}{1+\gamma} y_{it} \\ + \mathbb{I}_{h_{it}\neq 0}(1-\tau_t)\Pi_{it}^U + \mathbb{I}_{h_{it}=0}\tau_t^L (\Pi_t^E)^{1-\bar{\tau}^P} + Transfers_{it}, \end{aligned} \tag{46}$$

subject to

$$k_{it+1} \geq 0, \qquad b_{it+1} \geq \underline{B}. \tag{47}$$

A.3 Tax Revenue Construction

The average tax rate τ_t in the main text is the ratio of tax revenue to total pre-tax labor and entrepreneurial income. Let w_t denote the real wage, n_{it} hours worked, and h_{it} idiosyncratic productivity. The average tax rate is

$$\tau_t = \frac{\mathbb{E}_t\left[w_t n_{it} h_{it} + \mathbb{I}_{h_{it}=0}\Pi_t^E\right] - \tau_t^L \mathbb{E}_t\left[\left(w_t h_{it} n_{it} + \mathbb{I}_{h_{it}=0}\Pi_t^E\right)^{1-\tau_t^P}\right]}{\mathbb{E}_t[w_t n_{it} h_{it} + \mathbb{I}_{h_{it}=0}\Pi_t^E]}. \tag{48}$$

The level parameter τ_t^L adjusts to match the average tax rate implied by the tax rule in the main text.

Aggregate tax revenue is

$$Tax_t = \tau_t \mathbb{E}_t\left[w_t n_{it} h_{it} + \mathbb{I}_{h_{it}\neq 0}\Pi_{it}^U + \mathbb{I}_{h_{it}=0}\Pi_t^E\right], \tag{49}$$

where Π_t^U denotes union profits and Π_t^E denotes entrepreneurial profits.

A.4 Firm-Side Problem

A.4.1 Final Goods Producers

Final goods producers aggregate differentiated intermediate goods y_{jt} into final output Y_t using a CES aggregator:

$$Y_t = \left(\int y_{jt}^{\frac{\eta_t - 1}{\eta_t}} \, dj \right)^{\frac{\eta_t}{\eta_t - 1}}, \tag{50}$$

where η_t is the elasticity of substitution across varieties. The associated price index is

$$P_t = \left(\int p_{jt}^{1-\eta_t} \, dj \right)^{\frac{1}{1-\eta_t}}, \tag{51}$$

and demand for variety j is

$$y_{jt} = \left(\frac{p_{jt}}{P_t} \right)^{-\eta_t} Y_t. \tag{52}$$

A.4.2 Intermediate Goods Producers

Intermediate goods producers operate under monopolistic competition and produce with Cobb–Douglas technology:

$$y_{jt} = Z_t N_{jt}^{1-\alpha} (u_{jt} K_{jt})^{\alpha}, \tag{53}$$

where Z_t is total factor productivity, N_{jt} is labor, K_{jt} is capital, and u_{jt} is utilization.

Capital depreciation increases with utilization:

$$\delta(u_{jt}) = \delta_0 + \delta_1 (u_{jt} - 1) + \frac{\delta_2}{2} (u_{jt} - 1)^2. \tag{54}$$

Firm j minimizes

$$w_t^F N_{jt} + (r_t + q_t \delta(u_{jt})) K_{jt},$$

which yields

$$w_t^F = (1 - \alpha) \, mc_{jt} \, Z_t \left(\frac{u_{jt} K_{jt}}{N_{jt}} \right)^{\alpha}, \tag{55}$$

$$r_t + q_t\delta(u_{jt}) = u_{jt}\alpha\, mc_{jt}\, Z_t \left(\frac{N_{jt}}{u_{jt}K_{jt}}\right)^{1-\alpha}, \tag{56}$$

and

$$q_t\left[\delta_1 + \delta_2(u_{jt}-1)\right] = \alpha\, mc_{jt}\, Z_t \left(\frac{N_{jt}}{u_{jt}K_{jt}}\right)^{1-\alpha}. \tag{57}$$

Under symmetry, marginal costs are equalized across firms: $mc_{jt} = mc_t$.

Price setting is subject to Calvo frictions. Each period, a firm can reset its price with probability $1-\lambda_Y$. A reoptimizing firm maximizes the expected discounted value of profits:

$$\mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \lambda_Y^t \tau_t^L Y_t^{1-\tau_t^P} \left\{ \left(\frac{p_{jt}\bar{\pi}^t}{P_t} - mc_t\right) \left(\frac{p_{jt}\bar{\pi}^t}{P_t}\right)^{-\eta_t} \right\}^{1-\tau_t^P}. \tag{58}$$

The price Phillips curve is

$$\log\left(\frac{\pi_t}{\bar{\pi}}\right) = \beta\mathbb{E}_t\left[\frac{Y_{t+1}}{Y_t}\log\left(\frac{\pi_{t+1}}{\bar{\pi}}\right)\right] + \kappa_Y\left(mc_t - \frac{1}{\mu_t^Y}\right), \tag{59}$$

where

$$\kappa_Y = \frac{(1-\lambda_Y)(1-\beta\lambda_Y)}{\lambda_Y}, \qquad \frac{1}{\mu_t^Y} = \frac{\eta_t - 1}{\eta_t}. \tag{60}$$

A.4.3 Capital Goods Producers

Capital goods producers take the price of capital q_t as given and choose investment I_t to maximize

$$\mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t I_t \left\{ \Psi_t q_t \left[1 - \frac{\phi}{2}\left(\frac{I_t}{I_{t-1}} - 1\right)^2\right] - 1 \right\}, \tag{61}$$

where Ψ_t is the marginal efficiency of investment and ϕ governs investment adjustment costs.

The optimality condition for investment is, with $g_t^I = I_t/I_{t-1}$ and $S(g) = \phi(g-1)^2/2$,

$$1 = \Psi_t q_t[1 - S(g_t^I) - S'(g_t^I)g_t^I] \tag{62}$$

$$+ \beta\mathbb{E}_t\left[\Psi_{t+1}q_{t+1}S'(g_{t+1}^I)(g_{t+1}^I)^2\right]. \tag{63}$$

Capital evolves according to

$$K_{t+1} = (1 - \delta(u_t))K_t + \Psi_t[1 - S(g_t^I)]I_t. \tag{64}$$

A.4.4 Labor Packers and Unions

Households sell labor to unions, which supply differentiated labor services to labor packers. Labor packers aggregate these services into total labor input:

$$N_t = \left(\int \hat{n}_{jt}^{\frac{\zeta_t - 1}{\zeta_t}} \, dj \right)^{\frac{\zeta_t}{\zeta_t - 1}}, \tag{65}$$

where $\zeta_t > 1$ is the elasticity of substitution across labor types. Demand for labor type j is

$$\hat{n}_{jt} = \left(\frac{W_{jt}}{W_t^F} \right)^{-\zeta_t} N_t, \tag{66}$$

where W_{jt} is the nominal wage set by union j and W_t^F is the nominal wage paid by labor packers.

Unions face Calvo wage rigidity. Each period, a union can reset its wage with probability $1 - \lambda_w$. A reoptimizing union maximizes

$$\mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \lambda_w^t \frac{W_t^F}{P_t} N_t \left\{ \left(\frac{W_{jt} \bar{\pi}_W^t}{W_t^F} - \frac{W_t}{W_t^F} \right) \left(\frac{W_{jt} \bar{\pi}_W^t}{W_t^F} \right)^{-\zeta_t} \right\}. \tag{67}$$

The wage Phillips curve is

$$\log \left(\frac{\pi_t^W}{\bar{\pi}_W} \right) = \beta \mathbb{E}_t \left[\frac{w_{t+1}^F N_{t+1}}{w_t^F N_t} \log \left(\frac{\pi_{t+1}^W}{\bar{\pi}_W} \right) \right] + \kappa_w \left(mc_t^w - \frac{1}{\mu_t^W} \right), \tag{68}$$

where

$$\pi_t^W = \frac{W_t^F}{W_{t-1}^F} = \frac{w_t^F}{w_{t-1}^F} \pi_t, \qquad mc_t^w = \frac{w_t}{w_t^F}, \qquad \frac{1}{\mu_t^W} = \frac{\zeta_t - 1}{\zeta_t}, \tag{69}$$

and

$$\kappa_w = \frac{(1-\lambda_w)(1-\beta\lambda_w)}{\lambda_w}. \tag{70}$$

B Additional Calibration Details

This appendix reports the full steady-state calibration, implementation constants, policy-rule parameters, shock processes, and additional remarks on the parameters most closely tied to the mechanism.

B.1 Steady-State Calibration

I pin down the steady state by fixing a set of preference, technology, income-process, liquidity/portfolio, and fiscal parameters so that the model matches the same broad quarterly targets as Bayer et al. (2024). The targeted objects include earnings persistence and dispersion, worker–entrepreneur flows, the liquid–illiquid composition of household wealth, borrower prevalence, factor shares, government size, and tax progressivity.

Table 3: Steady-state calibration: parameters and targets (quarterly)

Block	**Symbol**	**Value**	**Target / Mapping**	**Data / Source**
Preferences and labor supply				
Risk aversion	ξ	4.000	Precautionary saving strength	Kaplan–Violante (2014)
Inverse Frisch elasticity	γ	2.000	Labor supply slope	Chetty et al. (2011)
Discount factor	β	0.98255	K/Y	NIPA long-run averages
Income process and occupation				

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Table 3: Steady-state calibration: parameters and targets (quarterly) *(continued)*

Block	**Symbol**	**Value**	**Target / Mapping**	**Data / Source**
Persistence (labor inc.)	ρ_h	0.980	Earnings persistence	Storesletten–Telmer–Yaron (2004)
Std. dev. (labor inc.)	σ_h	0.120	Earnings dispersion	Storesletten–Telmer–Yaron (2004)
$E \to W$ transition	ι	$1/16 = 0.0625$	Worker share in steady state	Guvenen–Kaplan–Song (2014)
$W \to E$ transition	ζ	$1/4500 \approx 2.22 \times 10^{-4}$	Upper-tail wealth concentration	WID/SCF
Technology and markups				
Capital share	α	0.318	Labor share ≈ 0.68	NIPA factor shares
Depreciation (quarterly)	δ_0	0.0215	K/Y	NIPA CFC
Price markup (gross)	μ^Y	1.10	Micro price-cost margins	Standard benchmark
Wage markup (gross)	μ^W	1.10	Micro wage markup	Standard benchmark
Liquidity, maturity, and asset supply				
Portfolio adj. prob.	λ	0.030	Liquid/illiquid turnover	SCF/NIPA (as in BBL)
Bond maturity	δ_B	0.200	Avg. duration $= 1/\delta_B$	Treasury duration stylized fact
Tradable profit share	$\omega_{\Pi}^{\text{ref}}$	0.150	Liquid equity-like paper	BBL steady-state mapping

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Table 3: Steady-state calibration: parameters and targets (quarterly) *(continued)*

Block	**Symbol**	**Value**	**Target / Mapping**	**Data / Source**
Retirement of profit shares	ι_{Π}^{ref}	0.016	Liquid claim turnover	BBL steady-state mapping
Fiscal block and normalizations				
After-tax coefficient	τ^L	0.645	$G/Y \approx 0.15$ in steady state	NIPA government share
Tax progressivity	τ^P	0.1022	Average progressivity	SOI
Gross inflation	π	1.000	Normalization (indexation)	Convention
Gross policy rate	R^{b}	1.000	Normalization (quarterly)	Convention
Borrower wedge	$\bar{R}$	0.0466	Borrower share $\sim 16\%$	SCF

Notes: The steady-state calibration targets the same broad long-run empirical objects as Bayer et al. (2024): income persistence and dispersion, worker–entrepreneur flows, the combined liquid paper supply ($Bgov + \bar{q}_{\Pi}$) relative to illiquid capital K, the prevalence of borrowers, factor shares, government size, and tax progressivity. "BBL steady-state mapping" refers to the accounting identity linking $(\omega_{\Pi}, \iota_{\Pi})$ to $\bar{q}_{\Pi}/Y$ in their steady state. Inflation and the gross policy rate are normalized under indexation. The borrower wedge is $\bar{R} = (1.20)^{1/4} - 1$.

B.2 Implementation Constants and Scenario Controls

A small number of parameters act as scenario controls or implementation constants. These objects do not discipline the steady-state targets in Table 3. The table distinguishes active scenario controls from legacy constants that do not govern the reported experiment.

Table 4: Implementation constants and scenario controls

Symbol	Value	Role	
ASHIFT	1.000	Legacy scale	Inactive in this experiment
$\omega_{\Pi}^{\text{bar}}$	0.20	Legacy reference	Effective mapping given in main text
TRANSFERtoG	1.00	Baseline Transfer/G ratio	Scenario toggle
BFRatio, GFRatio	0.95, 0.95	Funded allocation shares	Fiscal decomposition
α_G	0.0	Government spending in utility	Scenario toggle

B.3 Policy Rules and Shock Processes

This section reports the policy-rule parameters and background exogenous shock processes. The AR(1) innovations remain inactive in the one-time fiscal experiments. The funded and unfunded fiscal rules are the main policy objects for the paper's mechanism. The remaining shocks complete the quantitative environment.

Table 5: Policy-rule and shock-process parameters (calibrated baseline)

Block	Symbol	Value	Symbol	Value	Symbol	Value
Exogenous aggregate shocks						
Spread / wedge	ρ_A	0.90	σ_A	0.001		
TFP	ρ_Z	0.90	σ_Z	0.001		
Investment-good productivity	ρ_Ψ	0.90	σ_Ψ	0.001		
Price markup	ρ_{μ^Y}	0.90	σ_{μ^Y}	0.001		
Wage markup	ρ_{μ^W}	0.90	σ_{μ^W}	0.001		
Income-risk process						
Income-risk state	ρ_s	0.84	σ_s	0.001		
Cyclical sensitivity	Σ_N	0.00				
Monetary policy						

Continued on next page

Table 5: Policy-rule and shock-process parameters (calibrated baseline) *(continued)*

Block	**Symbol**	**Value**	**Symbol**	**Value**	**Symbol**	**Value**
Taylor rule	ρ_R	0.70	θ_π	0.50	θ_Y	0.125
Monetary shock	σ_R	0.001				
Fiscal policy: tax rules						
Avg. tax rate	ρ_τ	0.50	γ_B^τ	0.00	γ_Y^τ	0.00
Progressivity	ρ_P	0.50	γ_B^P	0.00	γ_Y^P	0.00
Progressivity shock	σ_P	0.001				
Fiscal components: purchases and transfers						
GF shock	$\rho_{GFshock}$	0.98	$\sigma_{GFshock}$	0.001		
GU shock	$\rho_{GUshock}$	0.98	$\sigma_{GUshock}$	0.001		
TF shock	$\rho_{TFshock}$	0.98	$\sigma_{TFshock}$	0.001		
TU shock	$\rho_{TUshock}$	0.98	$\sigma_{TUshock}$	0.001		
GF rule	ρ_{GF}	0.90	γ_{YGF}	0.50	γ_{BGF}	-0.50
GU rule	ρ_{GU}	0.90	γ_{YGU}	0.50	γ_{BGU}	0.00
TF rule	ρ_{TF}	0.90	γ_{YTF}	0.50	γ_{BTF}	-0.50
TU rule	ρ_{TU}	0.90	γ_{YTU}	0.50	γ_{BTU}	0.00

Notes: Values report the calibrated baseline used in the simulations. Government purchases and transfers are decomposed as $G_t = GU_t + GF_t$ and $Transfer_t = TU_t + TF_t$. Total long-term nominal government debt is decomposed as $Bgov_t = BFgov_t + BUgov_t$, where $BUgov_t \equiv Bgov_t - BFgov_t$. The rules for GF_t and TF_t feed back on $BFgov_t$, while the debt-feedback coefficients in the rules for GU_t and TU_t are zero. Negative coefficients on funded debt imply stabilizing future adjustment, while nonnegative coefficients on unfunded debt imply the absence of debt-stabilizing backing.

C Numerical Solution Methods

I obtain the local solution using a Reiter-style algorithm. This local step has four parts. I first compute the stationary equilibrium by solving the household problems and the invariant distribution jointly. I then reduce the dimensionality of the cross-sectional distribution. Next, I linearize the equilibrium conditions around the stationary equilibrium. Finally, I solve the resulting linear rational-expectations system by generalized Schur decomposition. All reported general-equilibrium experiments use this reduced first-order solution, with explicit one-time payment paths as described below.

C.1 Overview of the Computational Approach

Let the individual state be (m, k, s), where m denotes the liquid position, k denotes illiquid real capital, and s collects idiosyncratic income and occupation states. Aggregate states include nominal rigidities, fiscal and monetary policy variables, taxes, and the cross-sectional distribution $\Theta_t(m, k, s)$.

The computation proceeds as follows:

1. Construct grids for (m, k) and a Markov chain for s.

2. Solve the household block for adjusters and non-adjusters. Adjusters choose both m' and k'. Non-adjusters choose m' and keep k fixed.

3. Iterate the distribution forward using the household policy rules and the idiosyncratic transition matrix.

4. Reduce the distributional state, linearize the equilibrium system, and solve the linearized model using QZ/Schur methods.

C.2 Discretization: Idiosyncratic Risk and Asset Grids

Idiosyncratic states. Worker productivity follows a log-AR(1) process. I approximate it by a finite-state Markov chain with n_s states using a Tauchen-style method. Occupation switching between workers and entrepreneurs is incorporated by adding an entrepreneur state and modifying the transition matrix. Let $\Phi \in \mathbb{R}^{n_s \times n_s}$ denote the resulting transition matrix.

Liquid and illiquid grids. Liquid assets m are discretized on a non-uniform grid with more points near the borrowing constraint $\underline{m} < 0$. Illiquid capital k is discretized on a non-uniform grid with more points at low levels of capital and fewer points in the upper tail. Denote the grid sizes by n_m and n_k.

Portfolio adjustment. Each period, a fraction λ of households can adjust illiquid capital. With probability $1-\lambda$, a household keeps $k' = k$ and chooses only m'. The household block therefore yields two liquid-saving policies, $m'_a(m,k,s)$ and $m'_n(m,k,s)$, and one capital policy, $k'(m,k,s)$, for adjusters.

C.3 Household Problem and EGM Implementation

C.3.1 Recursive Formulation

Let $W_t(m,k,s)$ denote the continuation value before the capital-adjustment draw. The conditional value functions are

$$V_t^a(m,k,s) = \max_{m',k'} \; u(x_t) + \beta\, \mathbb{E}\left[W_{t+1}(m',k',s') \mid s\right], \tag{71}$$

$$V_t^n(m,k,s) = \max_{m'} \; u(x_t) + \beta\, \mathbb{E}\left[W_{t+1}(m',k,s') \mid s\right], \tag{72}$$

$$W_t(m,k,s) = \lambda V_t^a(m,k,s) + (1-\lambda) V_t^n(m,k,s), \tag{73}$$

subject to the budget constraint and the constraints on (m', k'). The choice variable entering utility is the GHH composite x_t.

C.3.2 Euler Equations and the EGM Step

For an interior liquid choice, the Euler equation is

$$u_x(x_t) \;=\; \beta\, \mathbb{E}_t\big[u_x(x_{t+1})\, R^m_{t+1}\big]\,, \tag{74}$$

where R^m_{t+1} is the real gross return on the liquid position, including the borrowing wedge when $m < 0$.

For a candidate next-period state (m', k', s'), define the expected marginal value of liquid wealth as

$$\mathcal{M}_t(m', k', s) \;:=\; \mathbb{E}\big[u_x(x_{t+1})\, R^m_{t+1} \mid s\big]\,. \tag{75}$$

Then (74) implies

$$u_x(x_t) = \beta\, \mathcal{M}_t(m', k', s) \quad \Rightarrow \quad x_t = u_x^{-1}(\beta\, \mathcal{M}_t(m', k', s))\,. \tag{76}$$

Non-adjusters. For non-adjusters, $k' = k$. Combining (76) with the budget constraint gives the endogenous current liquid position associated with each m':

$$m \;=\; \frac{m' + q_t k' + x_t - \text{income}_t(m, k, s)}{R^m_t}, \tag{77}$$

where $\text{income}_t(\cdot)$ collects labor income, profit income, transfers, and returns on k. I impose $m' \geq \underline{m}$ and use the standard EGM correction at the borrowing constraint.

Adjusters. For adjusters, the household chooses both m' and k'. The two-asset EGM finds liquid and illiquid positions that equalize their expected marginal continuation values at an interior optimum. For each candidate k', the liquid choice is obtained by interpolating

the zero of the marginal-value difference. The Euler equation then gives consumption, and the budget gives the endogenous resource grid. Constrained branches cover the borrowing limit and zero capital. Policies are interpolated back to the current state grid.

Interpolation. The EGM step produces policies on an endogenous grid. I map policies back to the exogenous (m, k) grid using monotone piecewise-linear interpolation.

C.4 Forward Iteration for the Cross-Sectional Distribution

Given policy functions and the idiosyncratic transition matrix Φ, the distribution evolves through a discrete transition operator. Let Θ_t denote the probability mass on the tensor grid $\mathcal{M} \times \mathcal{K} \times \mathcal{S}$. For each current grid point (m, k, s), the policy-implied next states are

$$(m'_a, k') \quad \text{for adjusters,} \qquad (m'_n, k) \quad \text{for non-adjusters.}$$

If a policy-implied state lies between grid points, I split mass across adjacent points using linear interpolation weights. I then apply the idiosyncratic transition matrix Φ and combine adjusters and non-adjusters:

$$\Theta_{t+1} = \lambda \, \mathcal{T}_a(\Theta_t) + (1 - \lambda) \, \mathcal{T}_n(\Theta_t),$$

where $\mathcal{T}_a$ and $\mathcal{T}_n$ are the transition operators implied by the two sets of policy functions.

Stationary distribution. The invariant distribution $\bar{\Theta}$ solves $\bar{\Theta} = \mathcal{T}(\bar{\Theta})$. I compute it by forward iteration with damping until the distribution is numerically stable and total mass is preserved.

C.5 Stationary Equilibrium

A stationary equilibrium consists of prices, policy variables, household policies, and an invariant distribution such that households optimize, firms and banks optimize, government budget identities and fiscal rules hold, and all markets clear.

Outer loop. I solve for the steady-state price and policy vector using a damped fixed-point or Broyden routine:

1. Guess the steady-state price and policy vector.
2. Solve the household block.
3. Compute the invariant distribution.
4. Aggregate household decisions and evaluate market-clearing residuals.
5. Update the price and policy vector.

Long-term nominal bond pricing. Long-term government debt has geometric maturity δ_B. In steady state, the no-arbitrage condition implies

$$\bar{q}^B = \frac{1}{\bar{R}^b - (1 - \delta_B)}. \tag{78}$$

I use this identity in the steady-state computation and as a check on the bond-pricing block.

C.6 Distribution Reduction for Linear Dynamics

The full tensor distribution $\Theta_t \in \mathbb{R}^{n_m n_k n_s}$ is too large to include directly in the linearized system. I therefore represent the distribution by its marginals and a compressed copula perturbation, using a discrete cosine transform (DCT). Marginal value functions are compressed separately.

DCT representation. Let θ denote a vectorized object on a grid of length n, and let $D \in \mathbb{R}^{n \times n}$ be the DCT matrix. The transformed coefficients are

$$\hat{\theta} = D\theta.$$

Low-frequency coefficients capture smooth movements in the distribution, while high-frequency coefficients capture local curvature near kinks and constraints.

Coefficient selection. For marginal value functions, coefficient selection retains a prescribed share of the ℓ_2 norm:

$$\frac{\|\hat{\theta}_{\mathcal{I}}\|_2}{\|\hat{\theta}\|_2} \geq 1 - \texttt{reduc_value}.$$

The retained marginal-value index set $\mathcal{I}$ is constructed by ordering coefficients by magnitude. Copula coefficients use their own reduction rule.

Reduced state vector. Let S denote the selection matrix that maps the full DCT coefficients into the retained subset:

$$\tilde{\theta} = S\hat{\theta}.$$

The reduced system combines these coefficients with the distributional marginals and aggregate variables. Household observables are differentiated using the reconstructed distribution and the household policy functions on the original grid, consistently with this reduced equilibrium system.

C.7 Linearization and Solution of the Rational-Expectations System

Stack the aggregate equilibrium conditions and the reduced distribution law into

$$\mathbb{E}_t \left[F \left(Z_{t+1}, Z_t, \varepsilon_{t+1} \right) \right] = 0, \tag{79}$$

where Z_t contains aggregate states, aggregate controls, and reduced distribution coefficients.

Linearizing around the steady state gives

$$A \, \Delta Z_{t+1} = B \, \Delta Z_t + C \, \varepsilon_{t+1}. \tag{80}$$

I solve (80) using generalized Schur decomposition. The solution yields policy functions of the form

$$\Delta Z_{t+1} = H \Delta Z_t + M \varepsilon_{t+1}, \qquad \Delta X_t = G \Delta Z_t,$$

where ΔX_t denotes selected controls.

C.8 Impulse Responses

For an unanticipated innovation in the local system, impulse responses can be computed by iterating

$$\Delta Z_{t+1} = H \Delta Z_t, \quad t \geq 1, \qquad \Delta Z_1 = M \varepsilon_1.$$

The current fiscal experiments instead impose the explicit additive payment path described in the main text, with a common total direct fiscal cost. Anticipated future payments enter households' information at date zero.

C.9 One-Time Payments in the Reduced First-Order System

I write the reduced linear equilibrium conditions in residual form as

$$A\,\Delta Z_{t+1} + D\,\Delta Z_t = f_t.$$

The initial predetermined states are at their stationary values. The terminal controls obey the stable policy function from the generalized Schur solution. This policy function is recomputed with both unfunded debt-feedback coefficients set to zero. The solution uses 320 quarters, and the reported paths are checked against 640 quarters.

The direct payment d_t is additive and lies outside fiscal-rule smoothing. For a component with steady-state value $\bar{x}$ and persistence ρ_x, its equation receives forcing $(d_t - \rho_x d_{t-1})/\bar{x}$. Thus a payment only at date zero does not mechanically generate a geometrically declining sequence of direct payments. Endogenous fiscal responses remain governed by the policy rule. Anticipated payments are announced at date zero and enter this same system at their specified arrival date.

For percentage plots, $d_0 = 0.05\bar{Y}$, using each economy's own quarterly steady-state output. Linear level deviations are obtained by multiplying the unit-payment solution by d_0; log deviations are not exponentiated. A targeted decile is defined before payment, splitting boundary atoms to obtain exactly ten percent of the population. Each recipient receives ten times the uniform payment. Its first-order forcing is the sum of the uniform-payment forcing and the automatically differentiated effect of a zero-sum redistribution at date zero.

Household budget components, consumption quantiles, and the consumption Gini are differentiated around the same steady state with respect to current and next-period reduced variables. These derivatives are evaluated along the first-order equilibrium path. Consumption quantiles use current liquid wealth and include changes in group composition; they do not follow fixed initial cohorts. All dynamic figures include dates zero through forty.

C.10 Diagnostics and Numerical Accuracy

The diagnostics distinguish equation accuracy from approximation accuracy:

1. **Linear equilibrium equations.** Across 35 experiments, the maximum equation residual is 1.71×10^{-13} for unit-payment forcing, or 2.41×10^{-14} at the plotted payment scale.

2. **Household aggregation and budgets.** The maximum relative difference between aggregated household consumption and the macroeconomic consumption path is 1.81×10^{-11}. The household budget identity error is below 8.0×10^{-13} per unit of direct payment, including the conditional adjuster and non-adjuster budgets.

3. **Horizon sensitivity.** Extending the solution from 320 to 640 quarters changes consumption over dates zero through forty by at most 4.94×10^{-9} of the longer-horizon peak response.

These checks verify the solution of the reduced linear system and consistency of its household observables. They do not establish convergence with respect to the asset grids or the number of retained distributional coefficients. Responses to a payment of 5 percent of quarterly output are first-order extrapolations; accuracy of a nonlinear solution at that payment size has not been established.

Implementation note. Stationary equilibria, household policies, automatic derivatives, and reduced equilibrium matrices are constructed in `Julia`. The one-time linear path solver, residual correction, horizon checks, and figures use `Python`. The code retains the original liquid-equity calibration and changes only the specified experimental parameters.

D Additional Results

These experiments retain liquid equity, $\lambda = 0.03$, and zero debt feedback in both unfunded fiscal components. The policy comparisons use the local solution described in Appendix C.10

and keep total direct payments fixed. Debt maturity is reported in Section 3.5.

D.1 Household Budgets by Adjustment Status

Non-adjusters keep capital fixed and respond through consumption and liquid saving. Per 100 of direct payment, their impact consumption rises by 3.88 under funded policy and 263.49 under unfunded policy; liquid net saving changes by 99.77 and -261.37, respectively. Capital net purchases are zero. For adjusters, consumption changes by 2.96 and 237.82, liquid net saving by 149.56 and -3745.25, and capital net purchases by -48.87 and 3509.56. The aggregate is the weighted average with weights 0.03 and 0.97. The large conditional portfolio responses of adjusters do not imply similarly large aggregate capital purchases.

D.2 Arrival Date of a Transfer Announced Today

Delaying a funded payment from today to quarters four and ten reduces consumption at announcement per unit of payment from 0.0385 to 0.0254 and 0.0181. This relationship is not globally monotone: a quarter-twenty payment gives 0.0184. Under unfunded policy, the corresponding announcement responses are 2.627, 2.610, 2.593, and 2.577. Much of the unfunded response occurs when the policy becomes known, even before the direct payment arrives.

A separate first-order household diagnostic holds prices, income, and taxes at their common stationary values. The bottom half of initial liquid wealth is more sensitive to delaying receipt: its consumption response at announcement falls from 0.0766 per unit of payment for immediate receipt to 0.0321 for receipt in quarter ten. The top half changes from 0.0225 to 0.0225. At quarter forty, the two values are 0.00475 and 0.01115. This fixed-price comparison isolates receipt timing at the household level; it is distinct from the funded–unfunded general-equilibrium experiment.

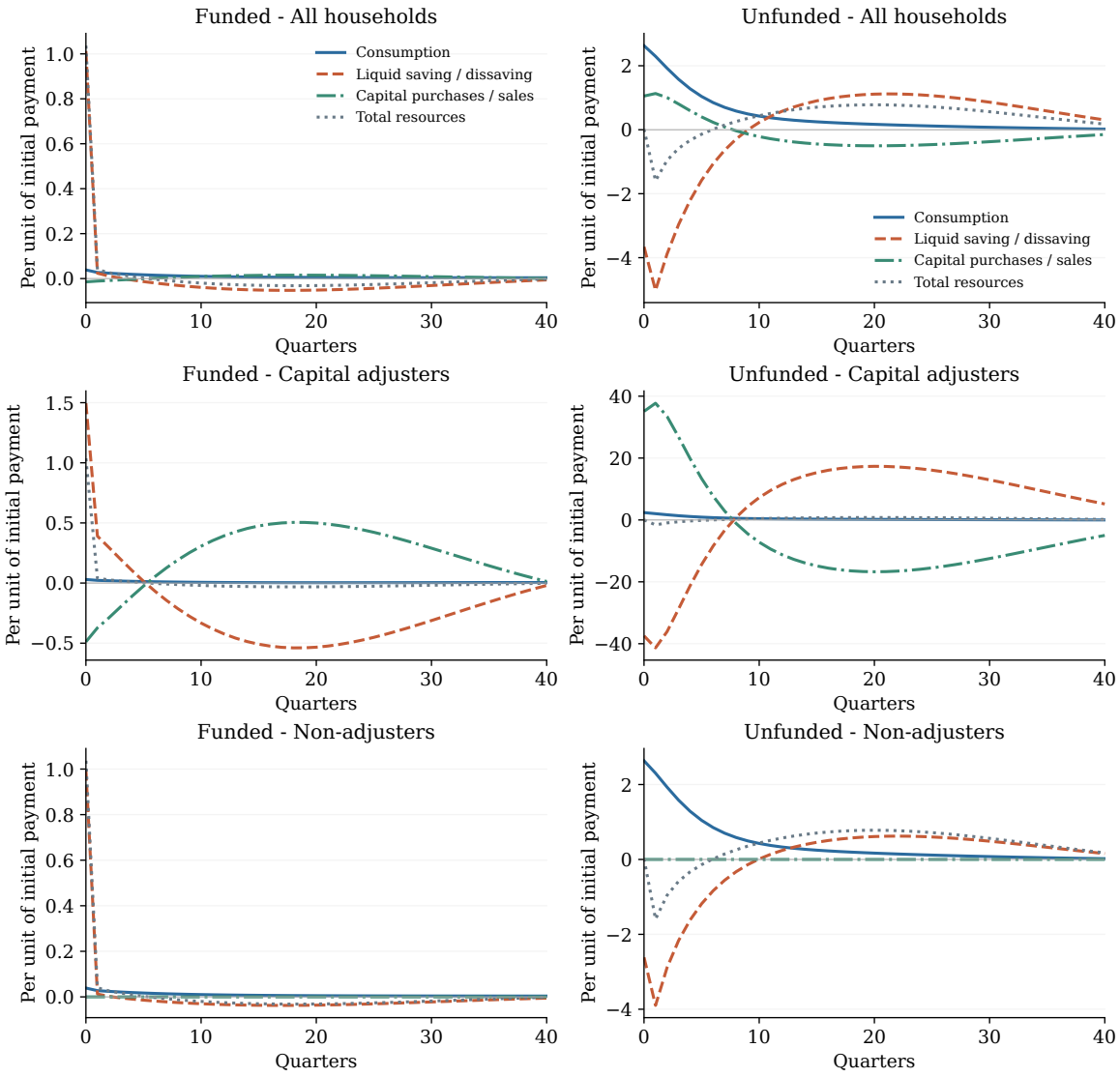


Figure 8: Household budget decomposition by adjustment status

Note: First-order changes relative to the stationary equilibrium per unit of the initial direct payment. Total resources include direct transfers and other resources. Non-adjusters make no capital trades. Positive and negative asset-flow responses denote changes in net saving and net purchases.

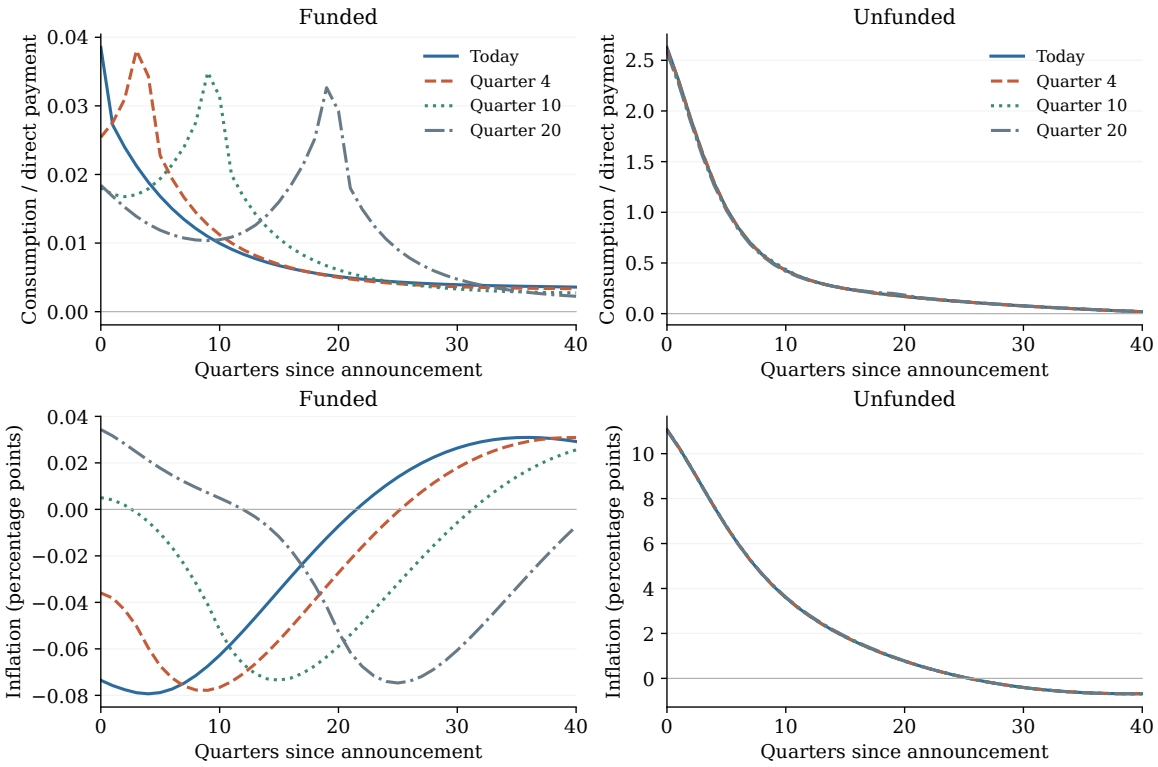


Figure 9: Arrival date of a transfer announced at date zero

Note: First-order responses; the same direct payment arrives at date zero, four, ten, or twenty. Consumption is divided by the payment; inflation is in percentage points for a payment of 5 percent of quarterly output. Vertical scales differ across regimes.

D.3 Policy-Rule Persistence

Changing transfer-rule persistence changes endogenous fiscal adjustment while the direct payment remains one-time. For persistence 0.50, 0.90, and 0.98, funded impact consumption per unit of payment is 0.0372, 0.0385, and 0.0383. Unfunded impact consumption remains approximately 2.627. With zero debt feedback, greater rule persistence does not prolong a debt-induced stream of unfunded transfers. The remaining output-growth feedback has little effect in this comparison.

D.4 Partial Fiscal Backing

Assigning a larger share of the same initial payment to the funded component reduces impact consumption and inflation. For funded shares of 0, 0.25, 0.50, 0.75, and 1, impact consumption per unit of payment is 2.627, 1.980, 1.333, 0.686, and 0.0385. Because this exercise uses the first-order solution, these mixtures are linear combinations of the endpoint responses.

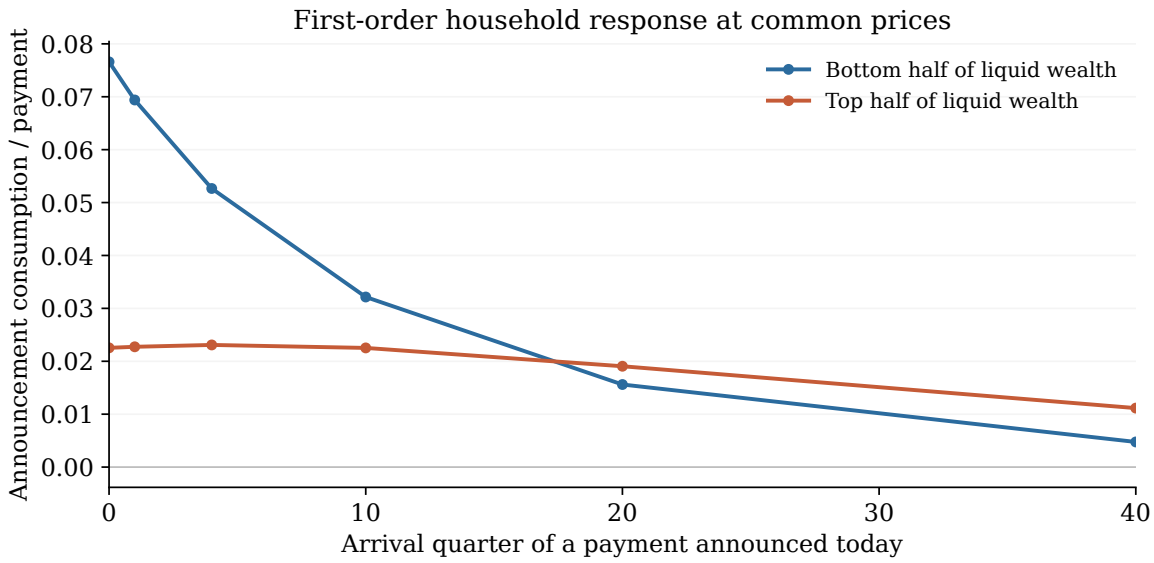


Figure 10: Liquid wealth and sensitivity to payment arrival at common prices

Note: Conditional mean consumption at announcement divided by the household payment. Households know at date zero when the uniform payment will arrive. The aggregate economy is not re-solved.

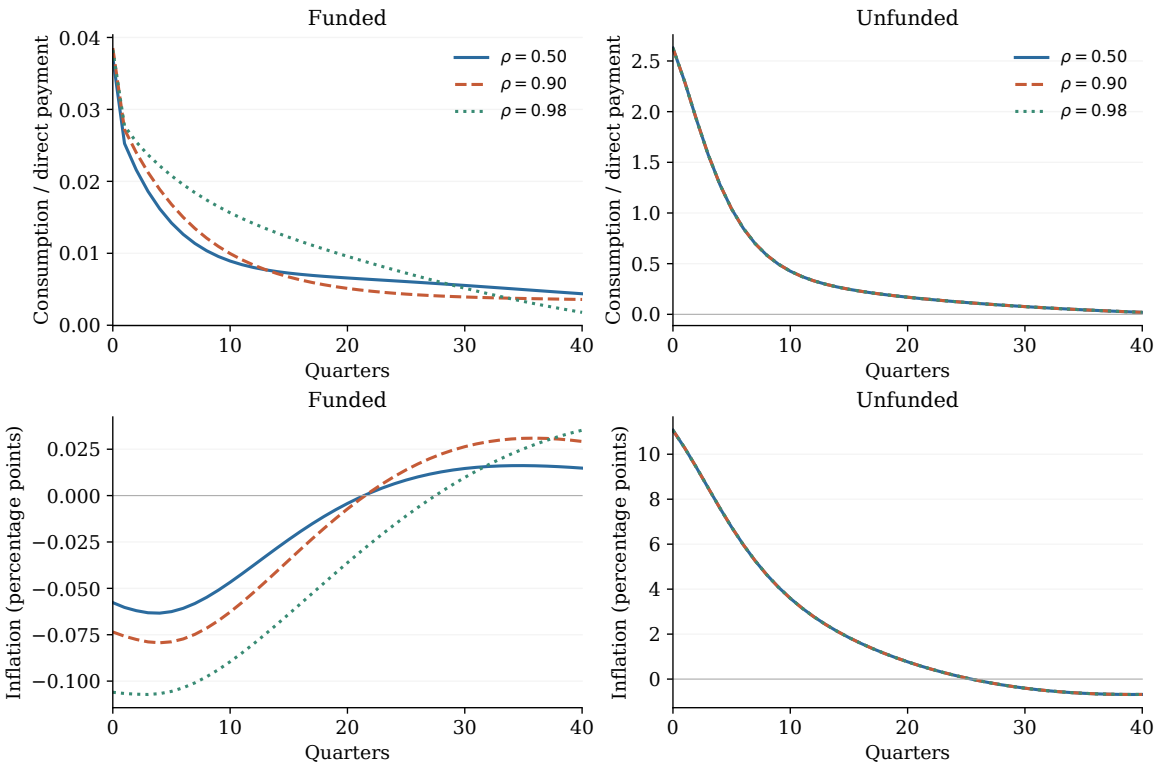


Figure 11: Persistence of the transfer policy rule

Note: Only the relevant transfer rule's persistence changes; the direct payment remains one-time and both unfunded debt-feedback coefficients remain zero. Units follow Figure 9.

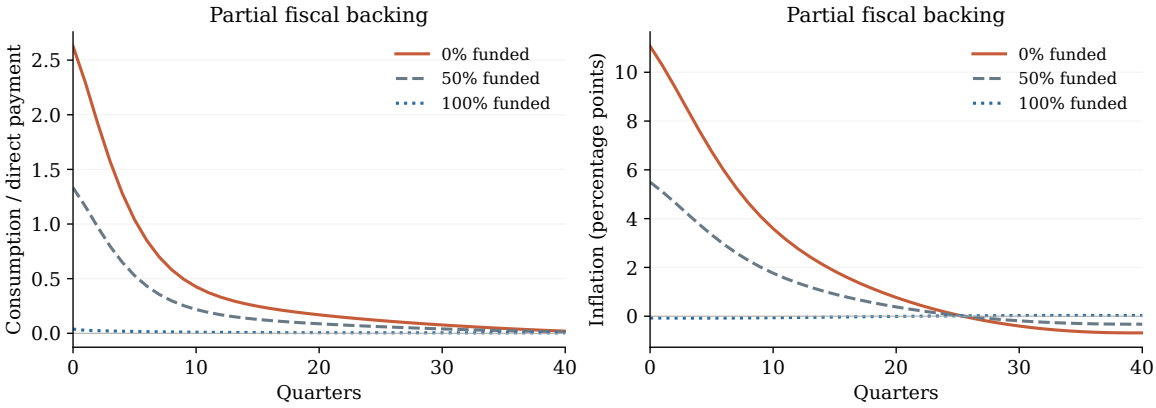


Figure 12: Partial fiscal backing of the same initial payment

Note: The figure shows fully funded, fully unfunded, and fifty-percent funded payments. All policy coefficients and total direct payments are held fixed. Units follow Figure 9.

E Robustness Checks

E.1 Government Spending in Utility

The benchmark retains $\alpha_G = 0$. The extension in which government spending enters household utility through $c + \alpha_G G$ remains a separate robustness exercise. The earlier figures for this extension used a different fiscal experiment and have not been recomputed for the one-time, zero-feedback specification. They are therefore not used as evidence for the results reported here.